

Molecular Biology of the Cell

Course Introduction + Chapters 1 & 2

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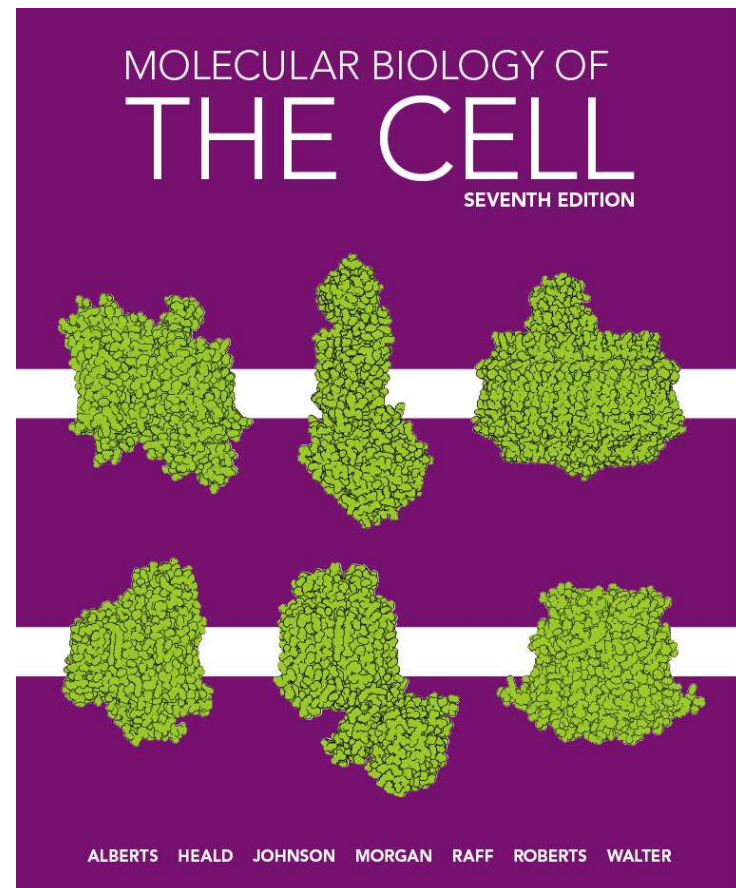
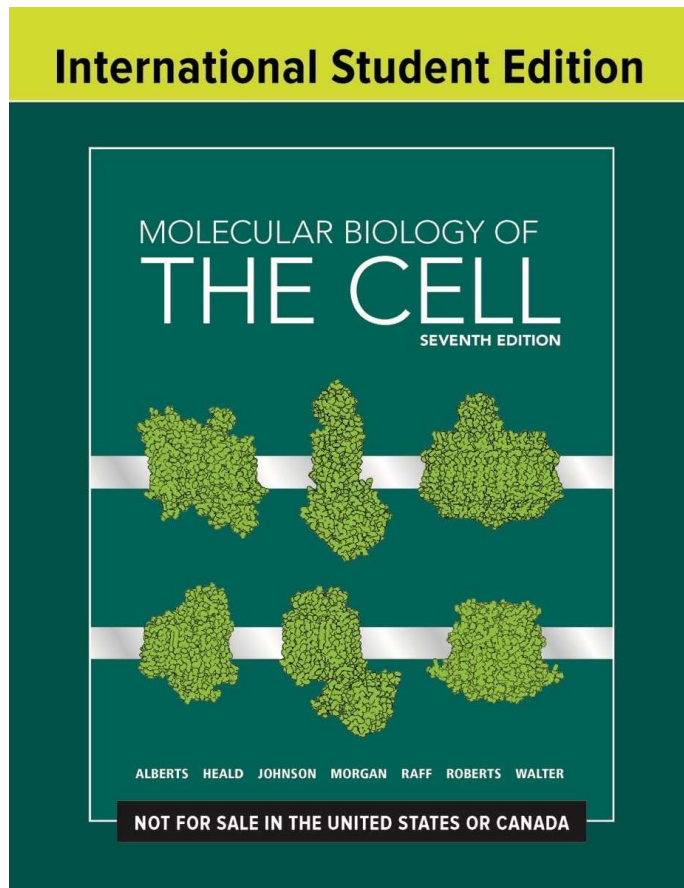
Thamar van Dijk

Ee1075

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Course Introduction

- refresh the memories
- level the playing field
- *'...and to deepen knowledge of molecular and cellular biology'*



Course Introduction

Book: 24 chapters

MBC-A: 8 lectures / chapters
Q&A session + exam

MBC-B: 8 lectures / chapters
Q&A session + exam

Final grade:

- average of A & B exams
- **average ≥ 5.5 to pass**

10-11 different lecturers

PART I

Chapter 1

Chapter 2

Chapter 3

PART II

Chapter 4

Chapter 5

Chapter 6

Chapter 7

PART III

Chapter 8

Chapter 9

PART IV

Chapter 10

Chapter 11

Chapter 12

Chapter 13

Chapter 14

Chapter 15

Chapter 16

Chapter 17

Chapter 18

PART V

Chapter 19

Chapter 20

Chapter 21

Chapter 22

Chapter 23

Chapter 24

INTRODUCTION TO THE CELL

Cells, Genomes, and the Diversity of Life

Cell Chemistry and Bioenergetics

Proteins

BASIC GENETIC MECHANISMS

DNA, Chromosomes, and Genomes

DNA Replication, Repair, and Recombination

How Cells Read the Genome: From DNA to Protein

Control of Gene Expression

WAYS OF WORKING WITH CELLS

Analyzing Cells, Molecules, and Systems

Visualizing Cells and Their Molecules

INTERNAL ORGANIZATION OF THE CELL

Membrane Structure

Small-Molecule Transport and Electrical Properties of Membranes

Intracellular Organization and Protein Sorting

Intracellular Membrane Traffic

Energy Conversion and Metabolic Compartmentation: Mitochondria and Chloroplasts

Cell Signaling

The Cytoskeleton

The Cell Cycle

Cell Death

CELLS IN THEIR SOCIAL CONTEXT

Cell Junctions and the Extracellular Matrix

Cancer

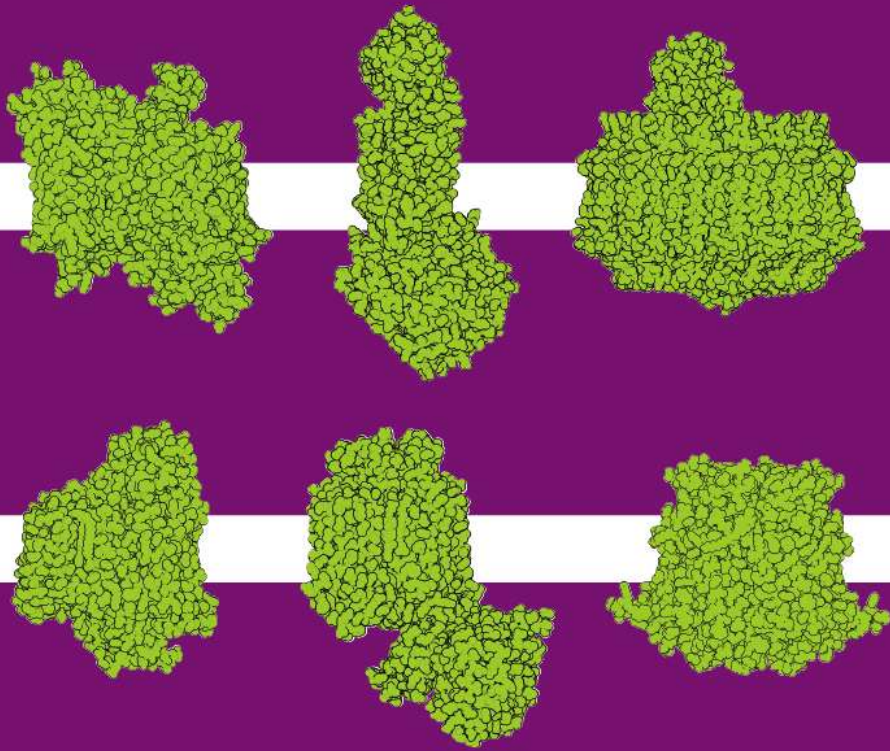
Development of Multicellular Organisms

Stem Cells in Tissue Homeostasis and Regeneration

Pathogens and Infection

The Innate and Adaptive Immune Systems

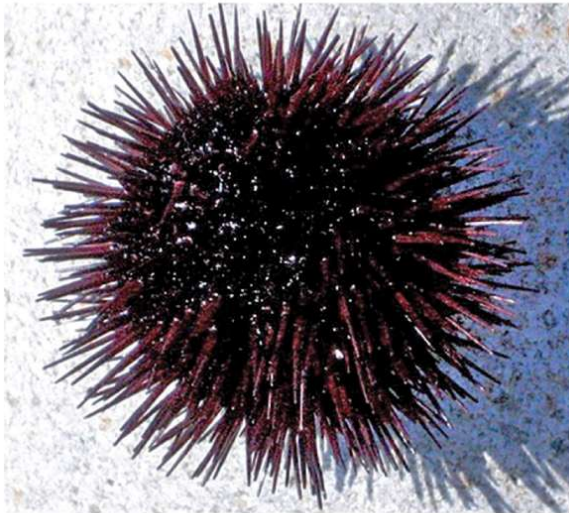
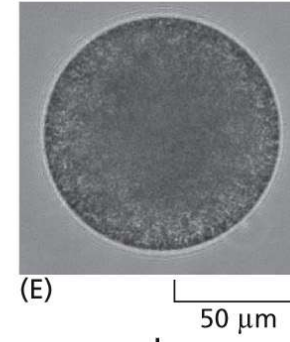
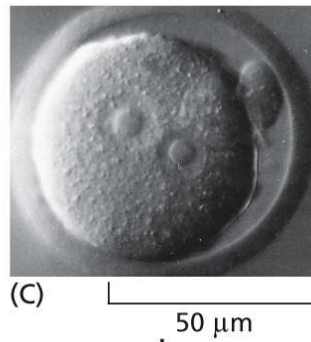
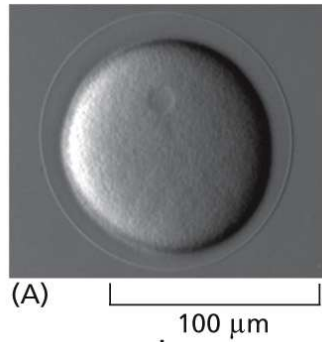
MOLECULAR BIOLOGY OF
THE CELL
SEVENTH EDITION



ALBERTS HEALD JOHNSON MORGAN RAFF ROBERTS WALTER

Chapter 1

Cells, Genomes, and the Diversity of Life



(B)



(D)



(F)

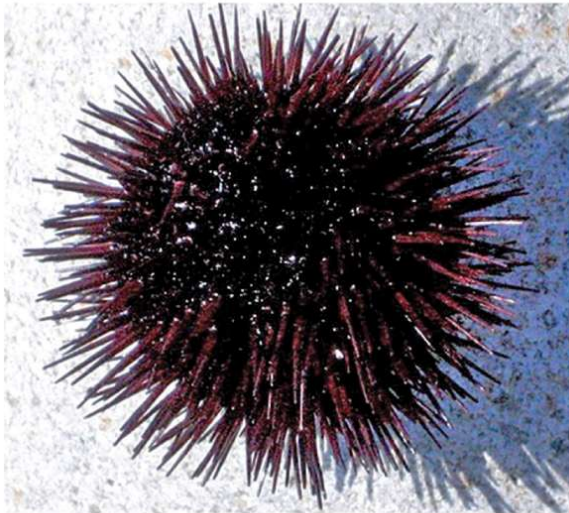
A, courtesy of David McClay; B, courtesy of Tim Hunt; C, courtesy of Patricia Calarco, from G. Martin, *Science* 209:768–776, 1980. With permission from AAAS; D, Rudmer Zwerver/Alamy Stock Photo; E and F, courtesy of Colin Brownlee

Cell: information + machinery (for interpretation and reproduction)

Human body: # of (human) cells?
of cell types?



(A) 100 μm



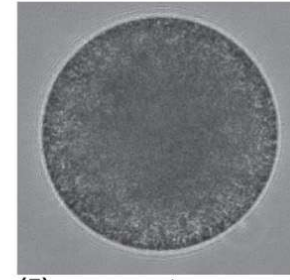
(B)



(C) 50 μm



(D)



(E) 50 μm



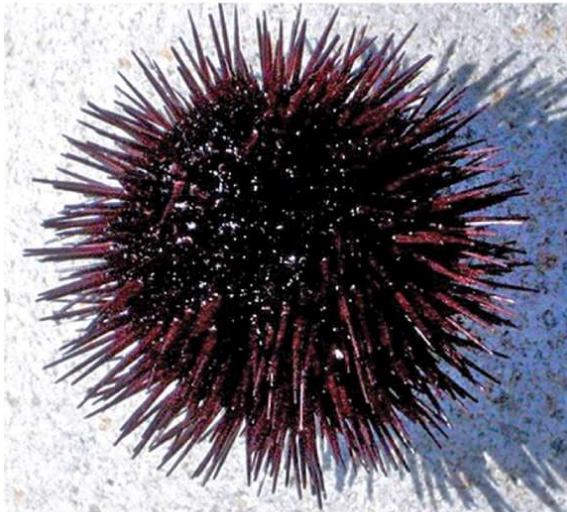
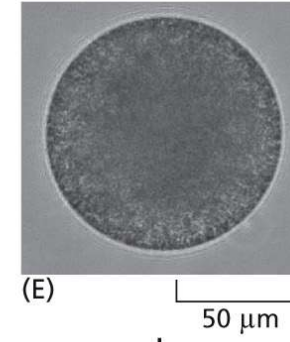
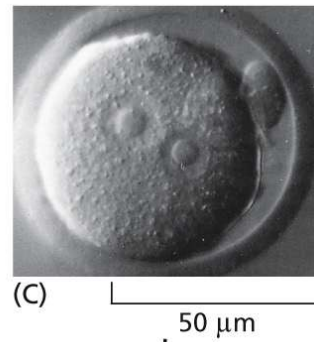
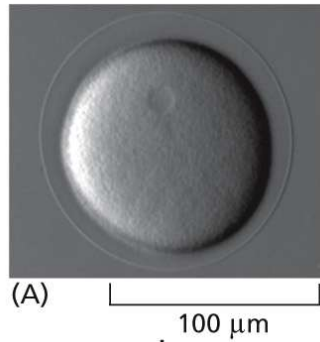
(F)

A, courtesy of David McClay; B, courtesy of Tim Hunt; C, courtesy of Patricia Calarco, from G. Martin, *Science* 209:768–776, 1980. With permission from AAAS; D, Rudmer Zwerver/Alamy Stock Photo; E and F, courtesy of Colin Brownlee

Cell: information + machinery (for interpretation and reproduction)

Human body: # of (human) cells?
of cell types?

$\sim 3 \times 10^{13}$



(B)



(D)



(F)

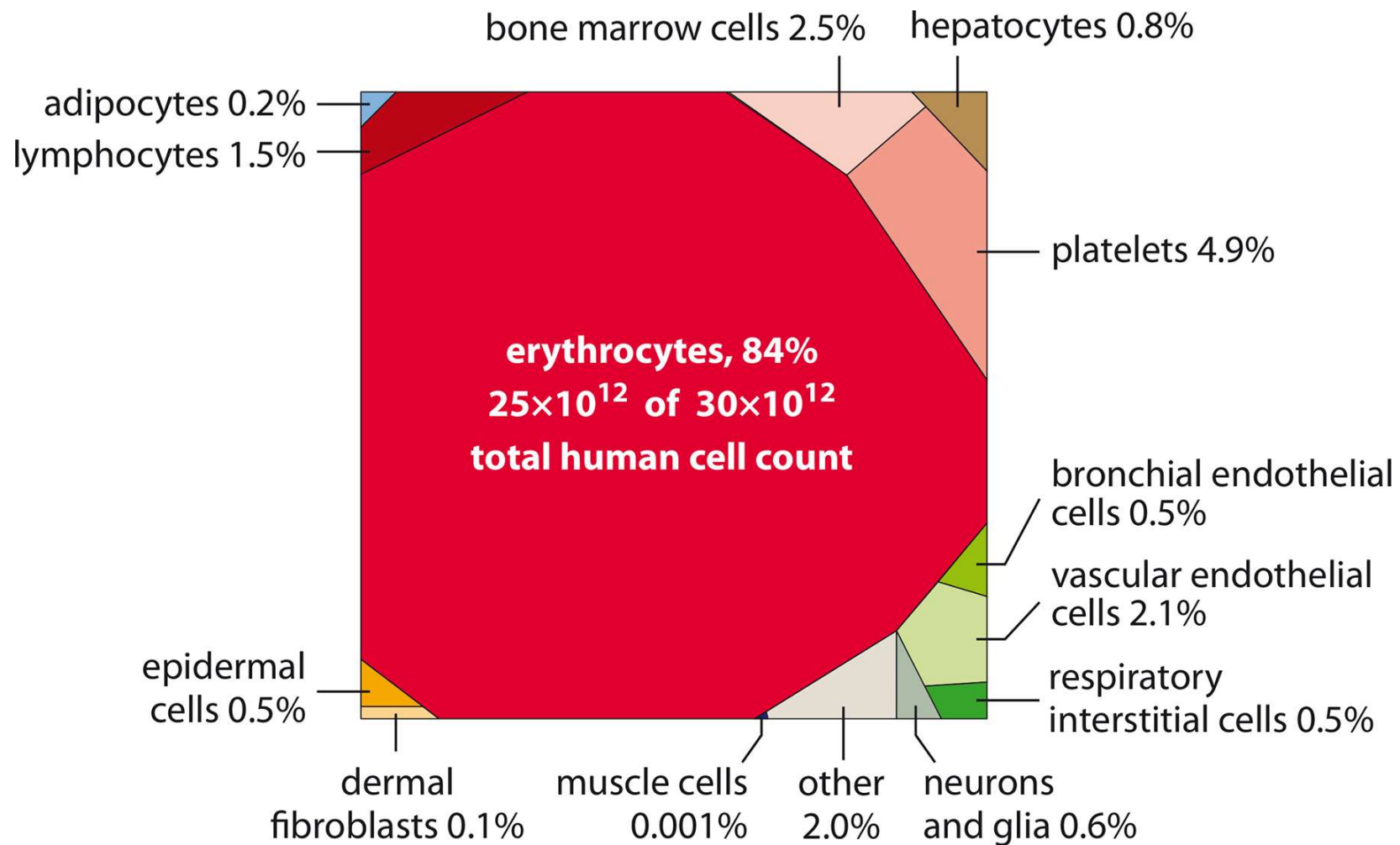
A, courtesy of David McClay; B, courtesy of Tim Hunt; C, courtesy of Patricia Calarco, from G. Martin, *Science* 209:768–776, 1980. With permission from AAAS; D, Rudmer Zwerver/Alamy Stock Photo; E and F, courtesy of Colin Brownlee

Cell: information + machinery (for interpretation and reproduction)

Human body: # of (human) cells?
of cell types?

$\sim 3 \times 10^{13}$

~ 325

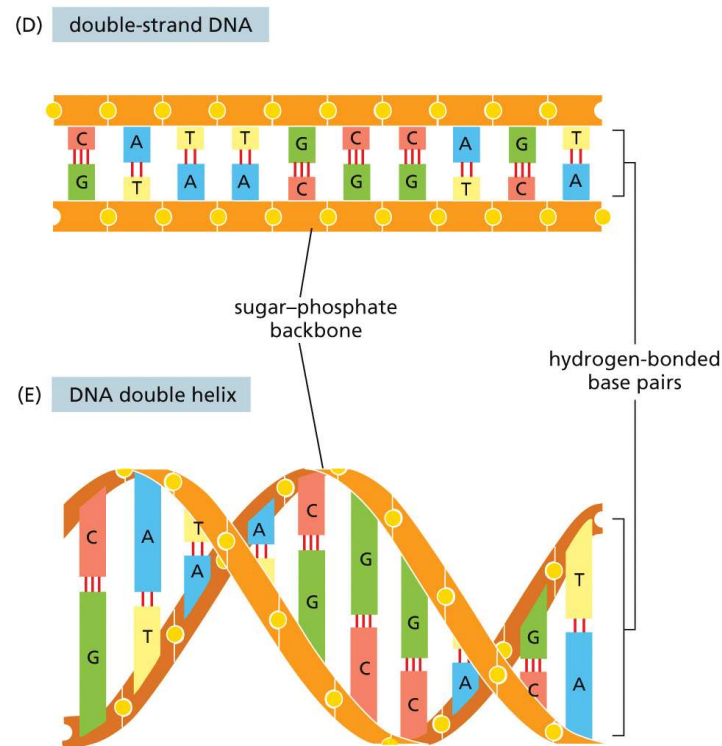


The Universal Features of life on Earth

- All Cells Store Their Hereditary Information in the Form of Double-Strand DNA Molecules

The Universal Features of life on Earth

- All Cells Store Their Hereditary Information in the Form of Double-Strand DNA Molecules



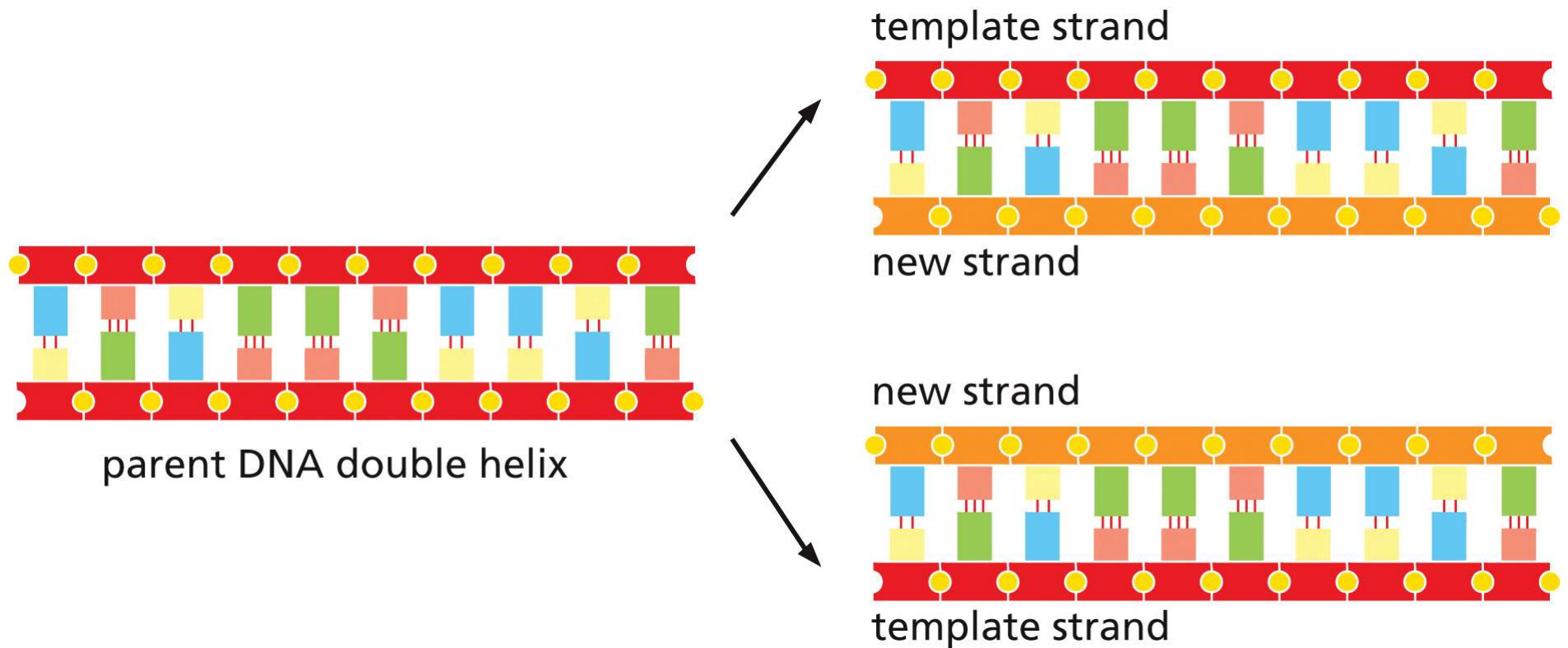
Adenine, Guanine, Cytosine, and Thymine

The Universal Features of life on Earth

- All Cells Duplicate Their Hereditary Information by Templated Polymerization

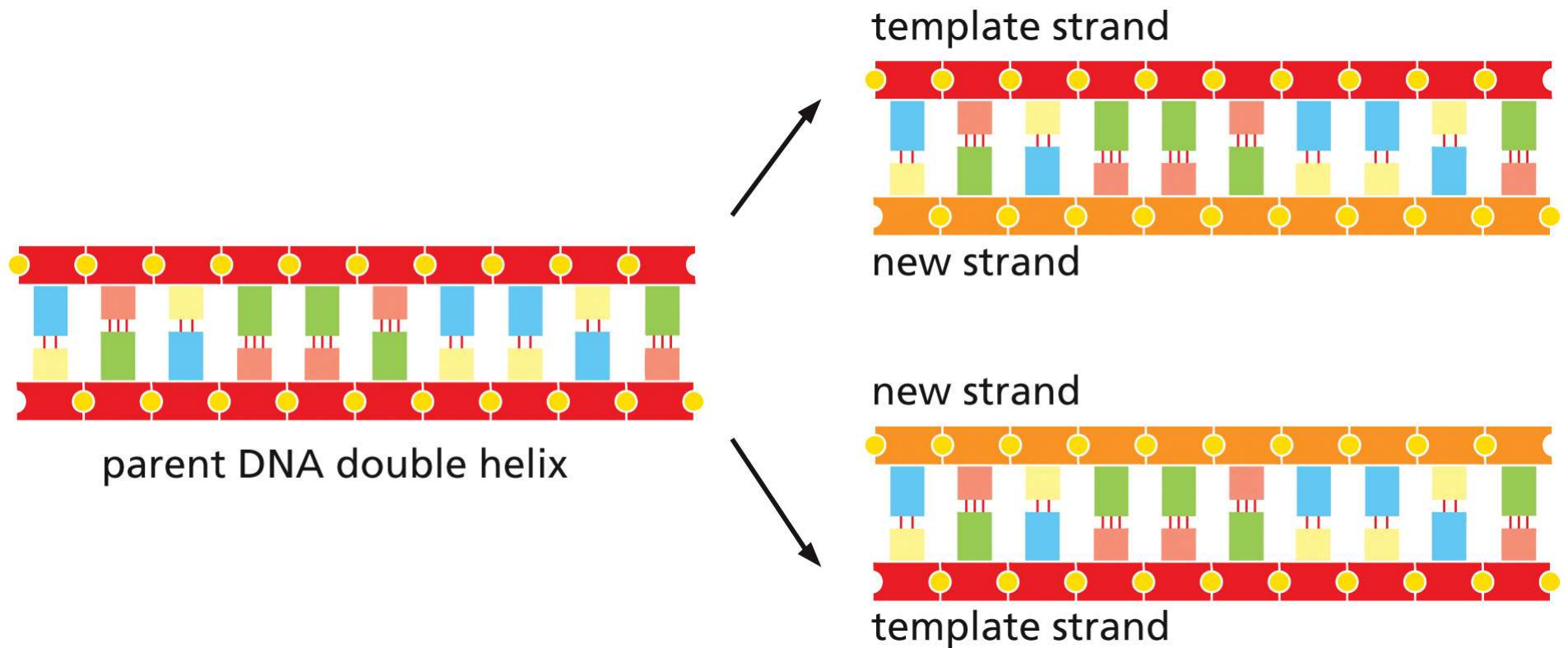
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The Universal Features of life on Earth

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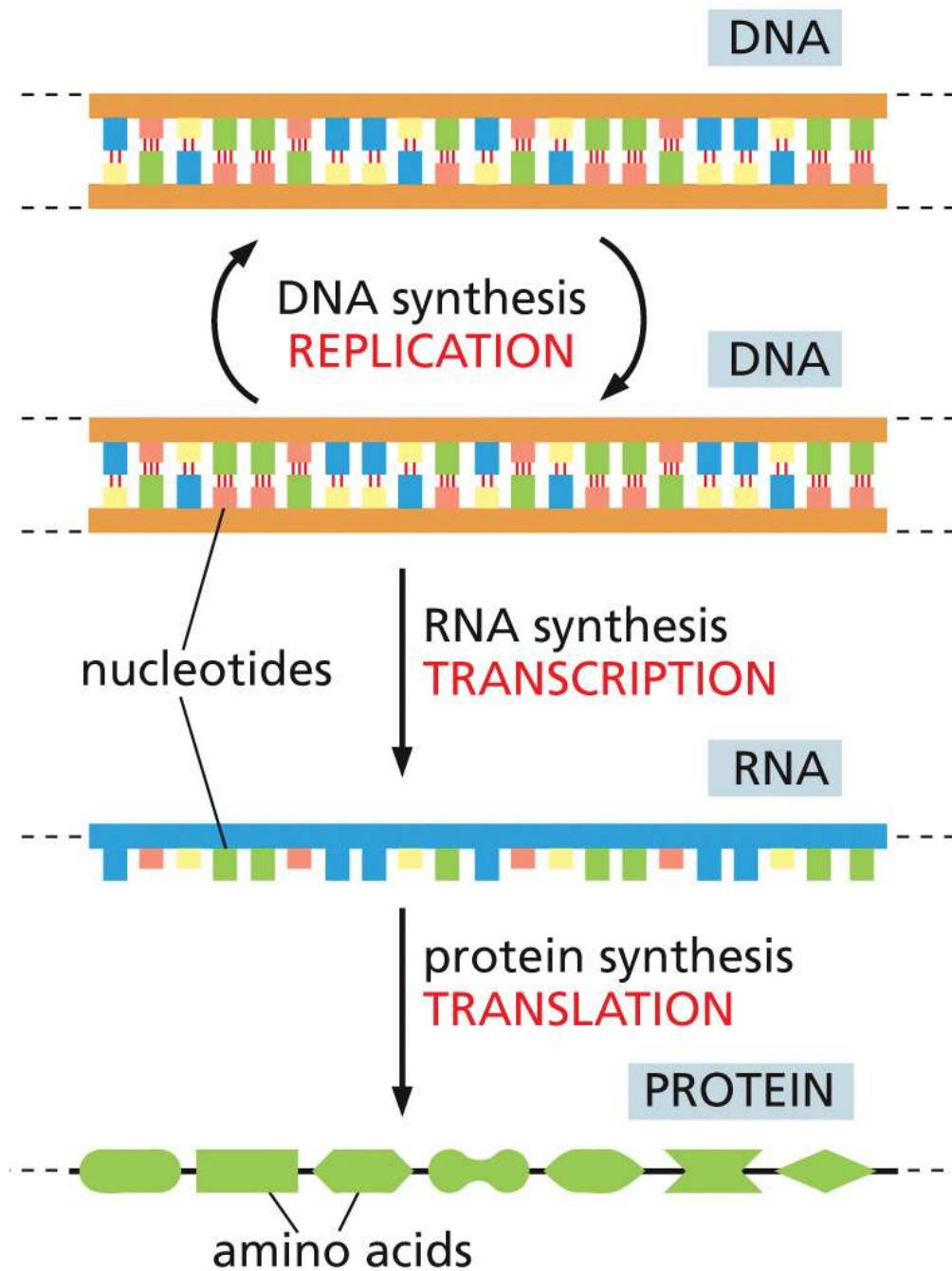


Copying genetic information by **replication**

The Universal Features of life on Earth

- All Cells Transcribe Portions of Their DNA into RNA Molecules

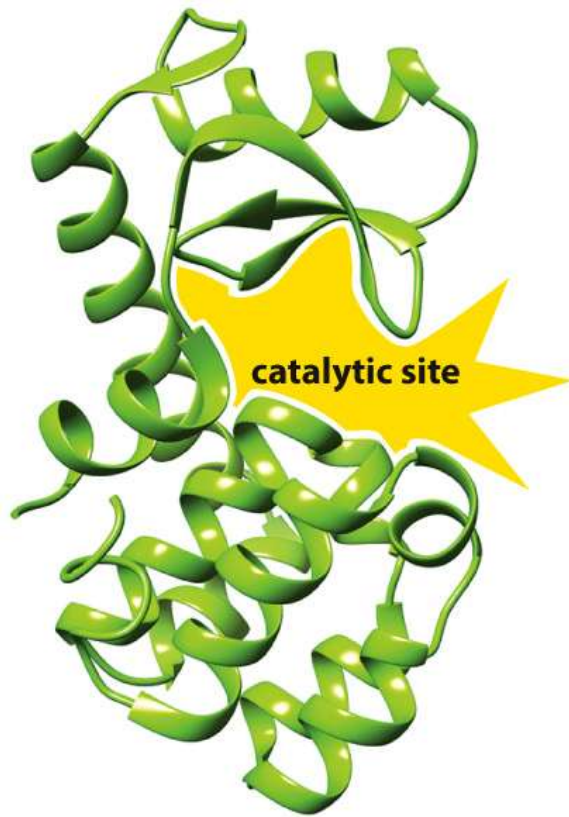
From DNA to protein



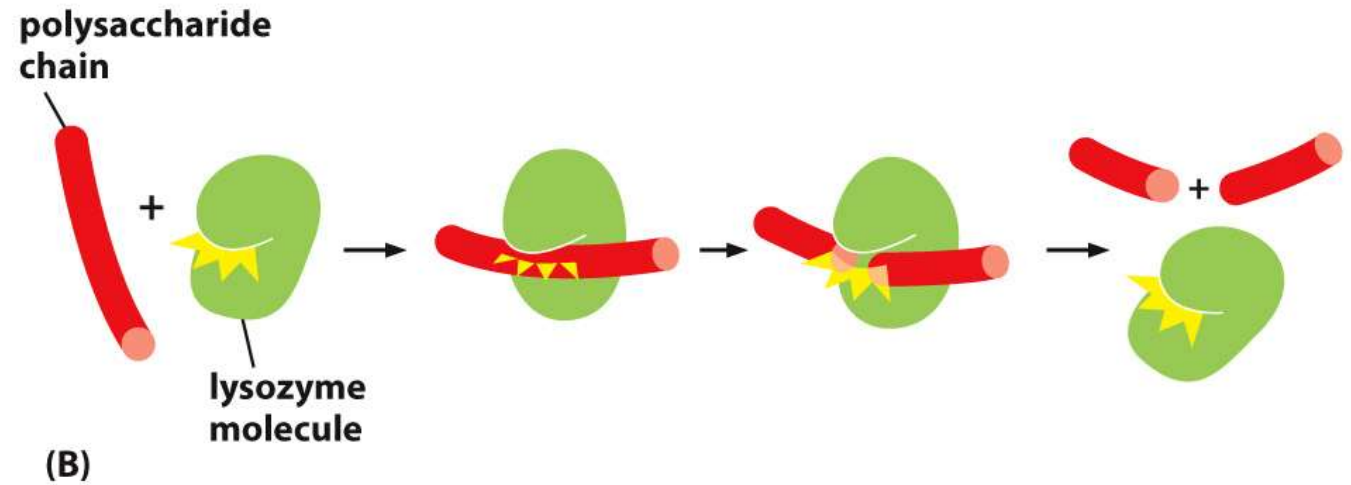
The Universal Features of life on Earth

- All Cells Use Proteins as Catalysts

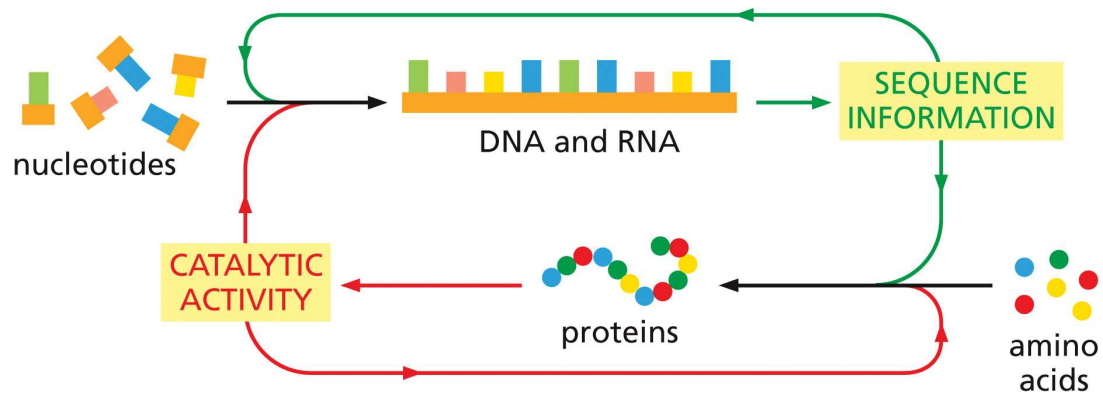
Protein: specific shape dictated by its sequence



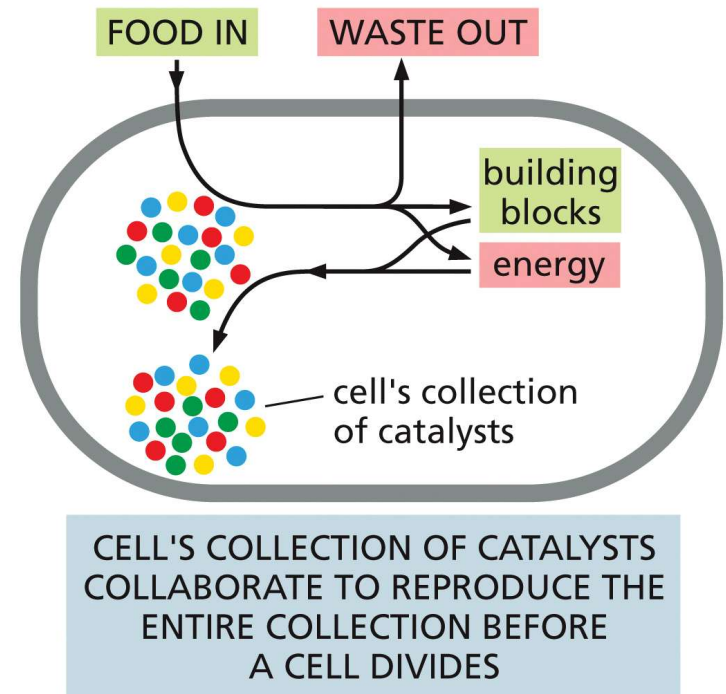
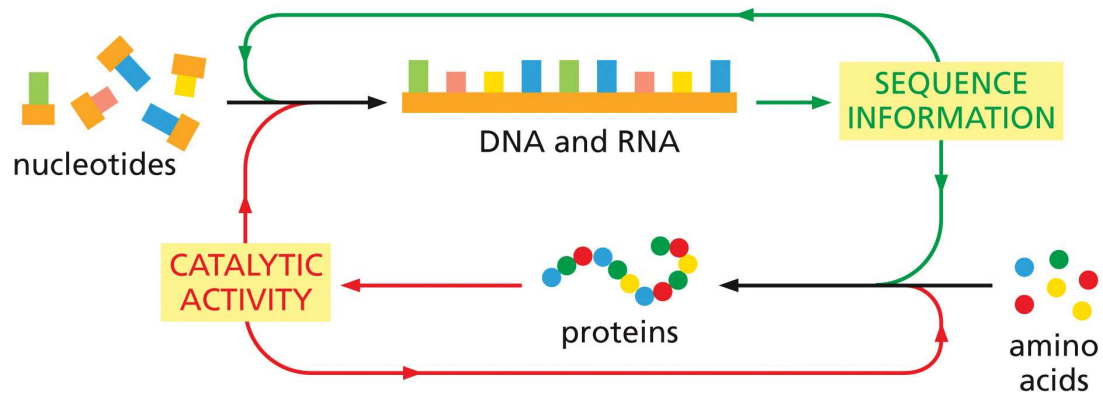
(A) lysozyme



Life as an autocatalytic process



Life as an autocatalytic process



The cell as a self-replicating collection of catalysts

Formation of a (plasma)membrane: phospholipids

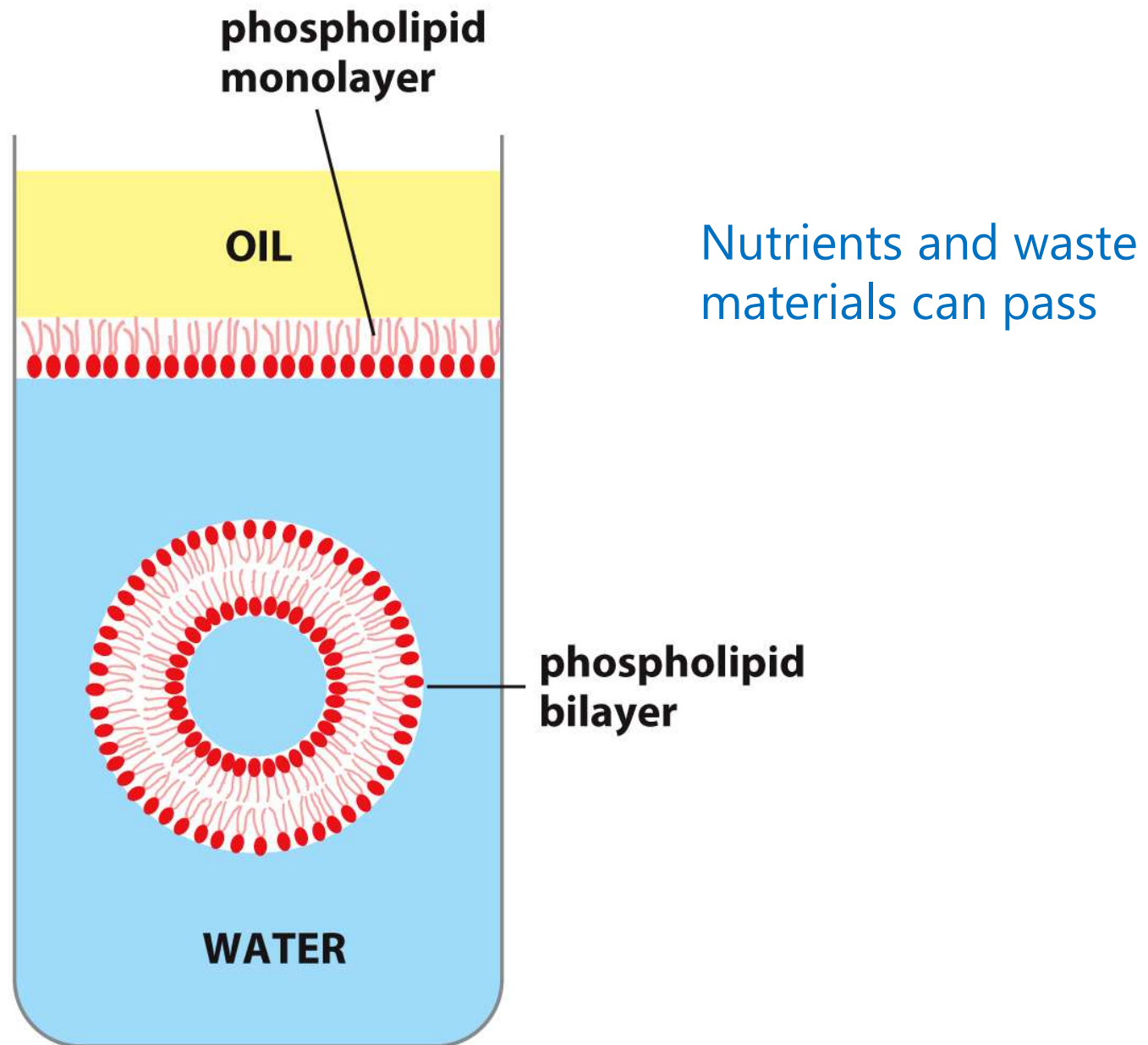
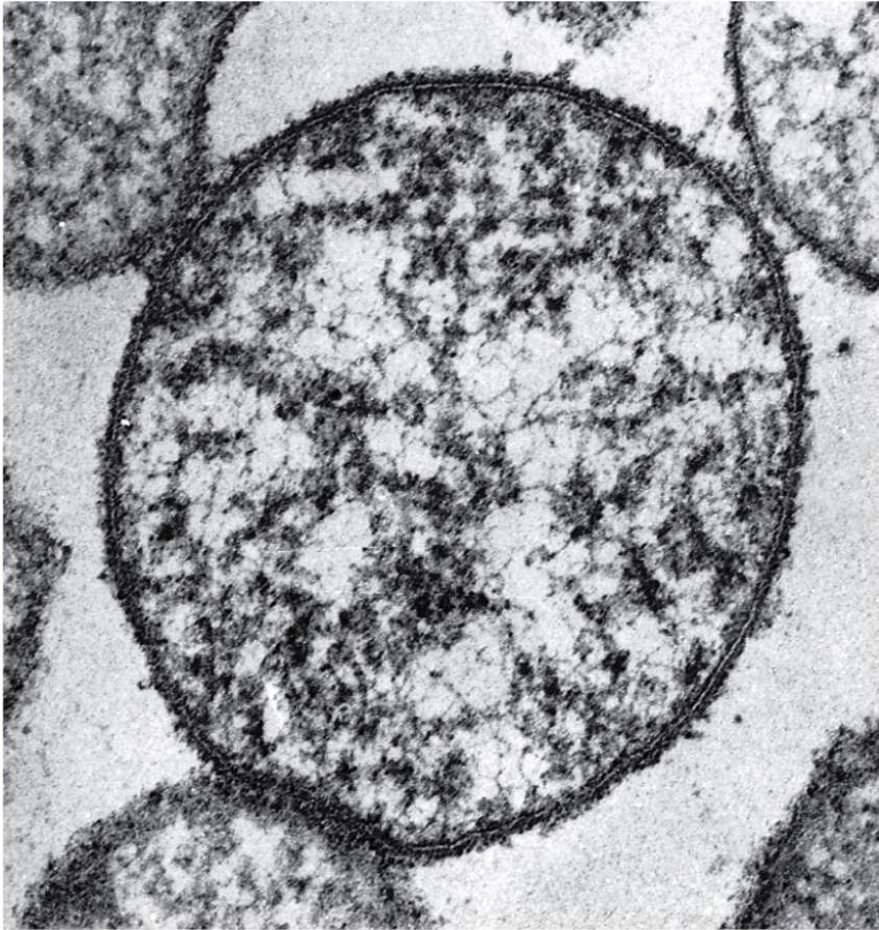


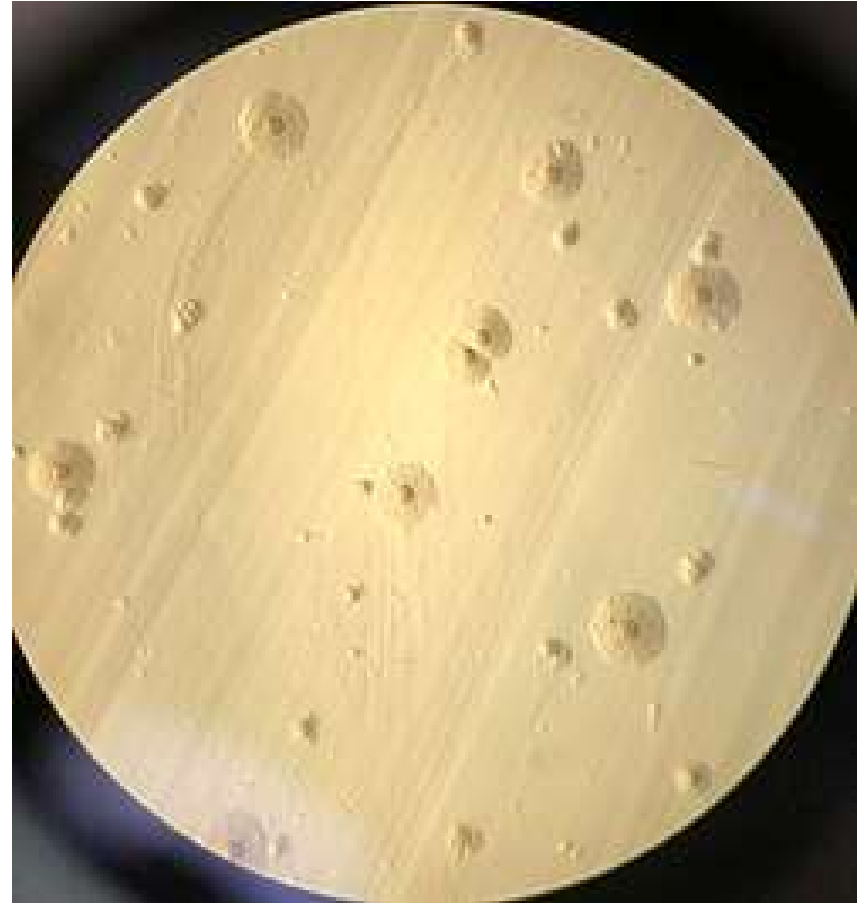
Figure 1-9 Molecular Biology of the Cell 6e (© Garland Science 2015)

Mycoplasma pneumoniae / *genitalium* / *laboratorium*



Courtesy of Roger Cole, in *Medical Microbiology*, 4th ed. [S. Baron, ed.], Galveston: University of Texas Medical Branch, 1996.

0.2 μm



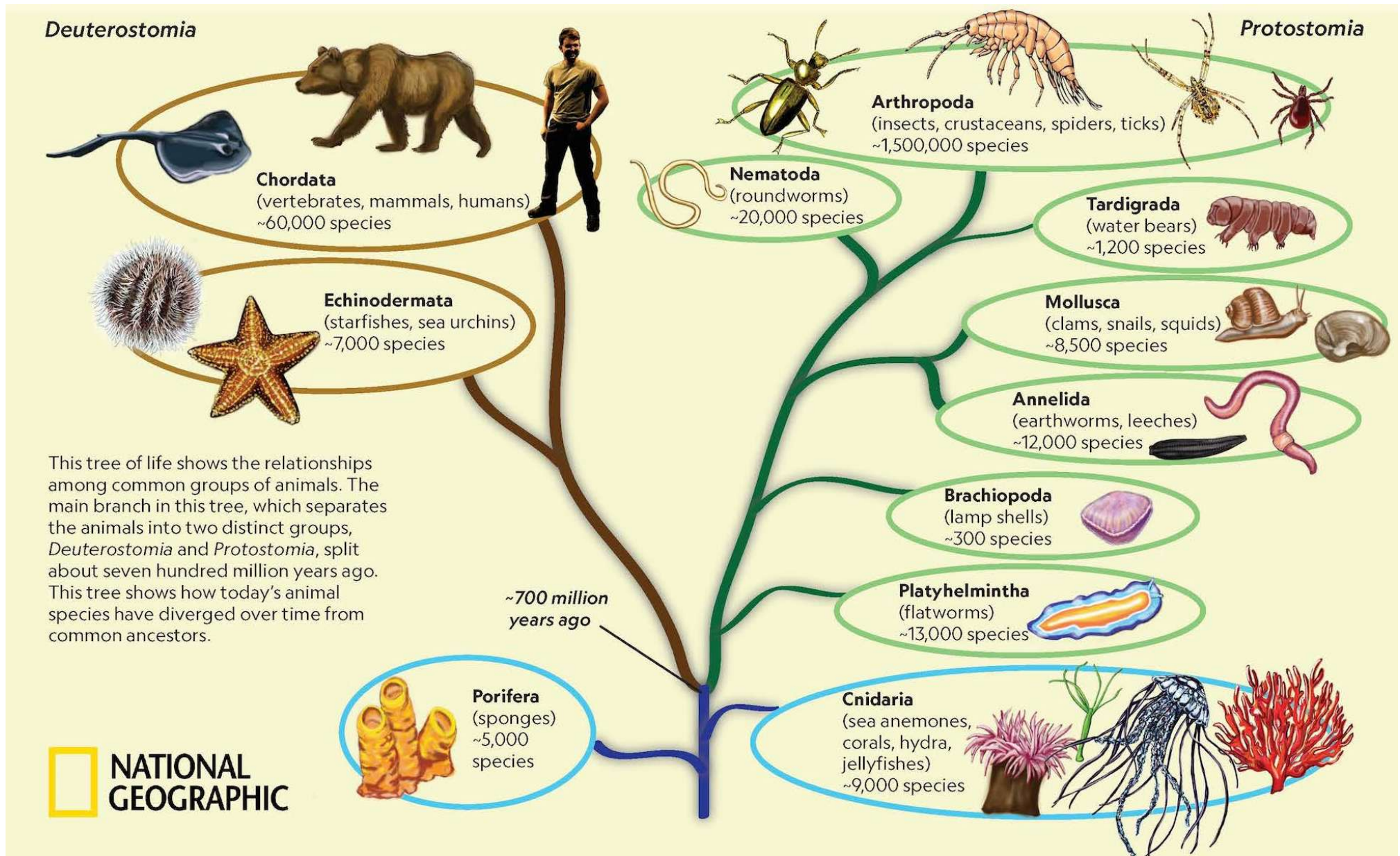
of genes?

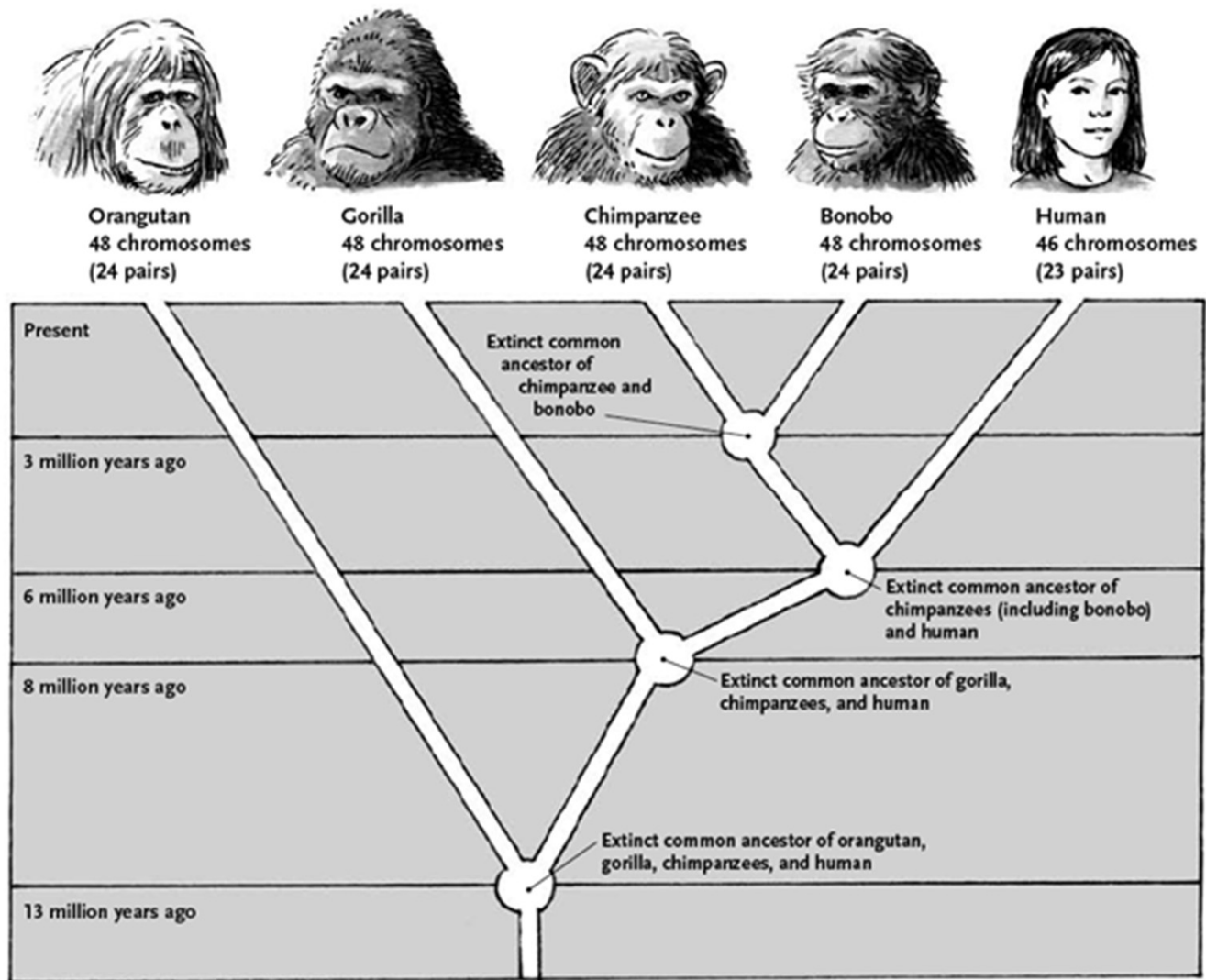
Summary so far ...

- individual cell is the minimal self-reproducing unit of living matter
- ... is a self-replicating collection of catalysts
- Transmission of information to progeny: DNA
- Proteins are the catalysts for most chemical reactions
- Function of protein is determined by amino acid sequence, specified by DNA sequence
- So genome determines the biochemistry of a cell
- The biochemistry of every living cell is fundamentally similar
- Chemistry of DNA/RNA/proteins (life) requires ~500 genes

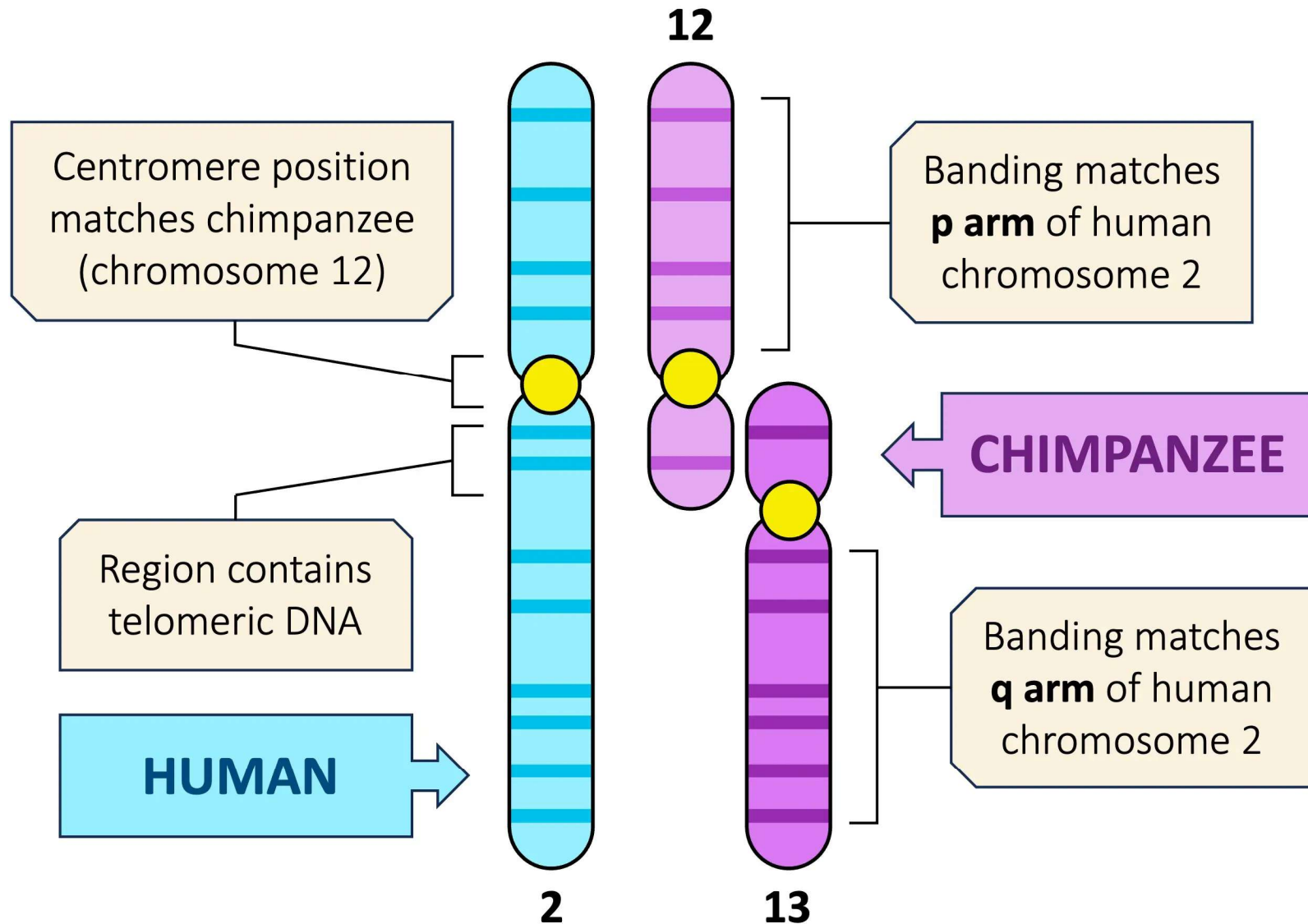
The tree of life, a classical view

based on: evolutionary relationships (developmental)

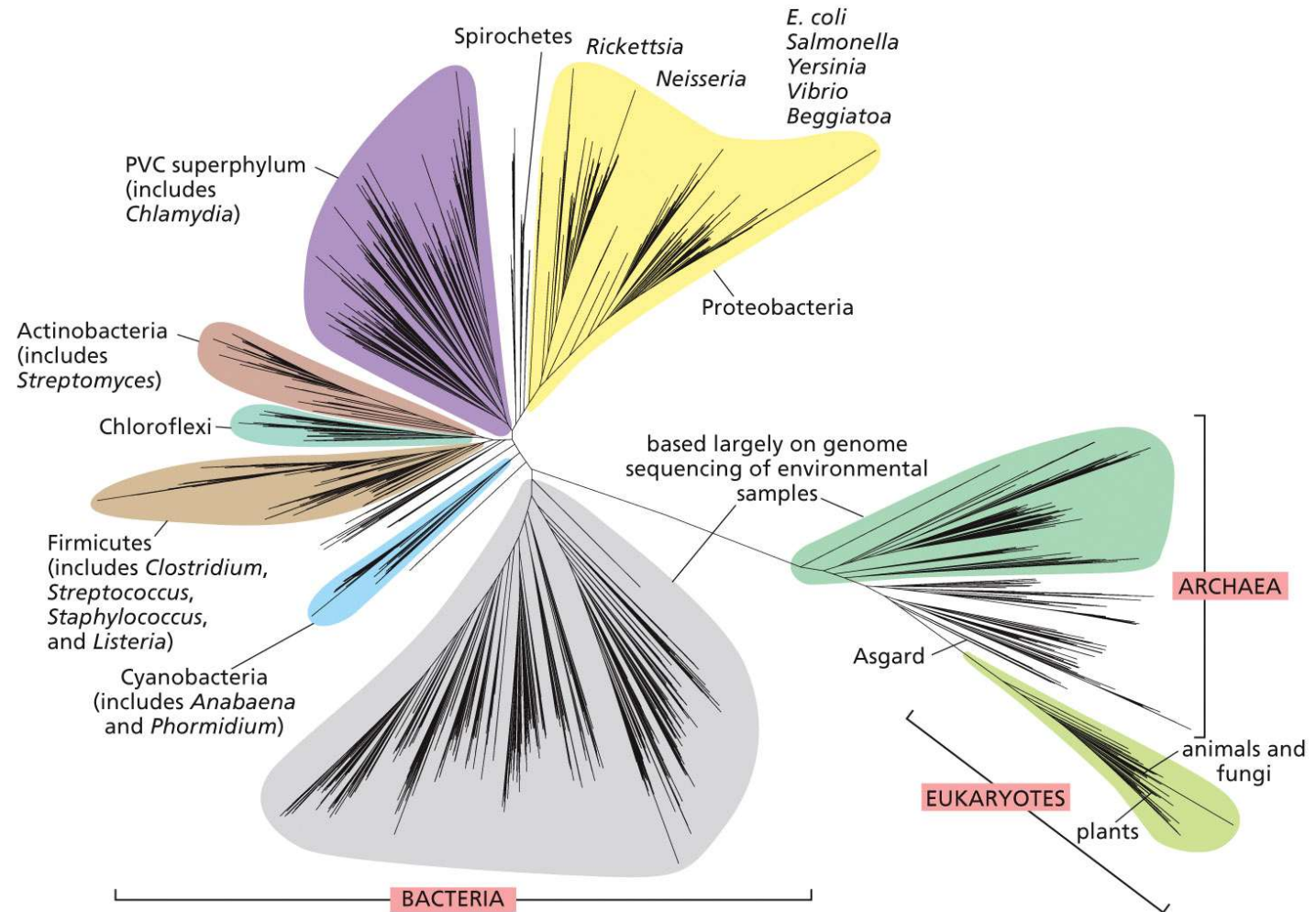




https://www.pbs.org/wgbh/nova/teachers/activities/3416_id_02.html



The tree of life: 3 domains

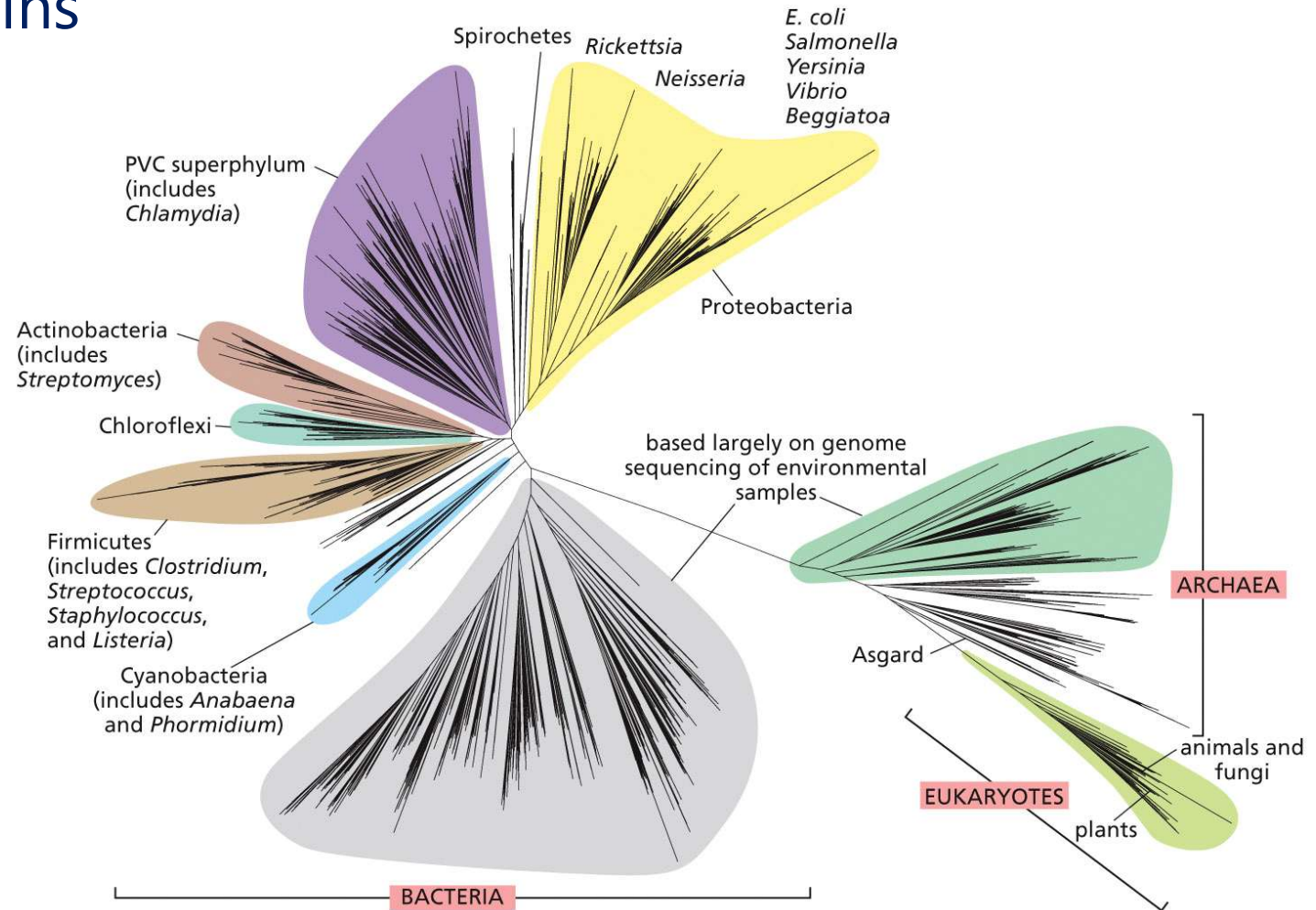


Adapted from C.J. Castelle and J.F. Banfield, *Cell* 172:1181–1197, 2018.

Archaeobacteria: '*intermediates*' between eubacteria and eucaryotes

They resemble eucaryotes on genetics, eubacteria on metabolism

The 3 domains



Adapted from C.J. Castelle and J.F. Banfield, *Cell* 172:1181–1197, 2018.

Tree of life, relationships based on ...?

Genome Diversification and the Tree of Life

- Genomes Diversify Over Evolutionary Time, Producing New Types of Organisms

GTTCCGGGGGAGTATGGTTGCAAAGCTGAAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human
GCCGCCTGGGGAGTACGGTCGCAAGACTGAAACTTAAAGGAATTGGCGGGGAGCACTACAACGGGTGGAGCCTGCGGTTTAATTGGATTCAACGCCGGGCATCTTACCA	<i>Methanococcus</i>
ACCGCCTGGGGAGTACGGCCGCAAGGTAAAACTCAAATGAATTGACGGGGGCCCGC•ACAAGCGGTGGAGCATGTGGTTTAATTCGATGCAACGCGAAGAACCTTACCT	<i>E. coli</i>
GTTCCGGGGGAGTATGGTTGCAAAGCTGAAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human

Part of rRNA component of ribosome
(after 3.5 billion years of evolution)

Bacteria: 1000-6000 genes

We?

GTTCCGGGGGAGTATGGTTGCAAAGCTGAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human
GCCGCCTGGGGAGTACGGTCGCAAGACTGAACTTAAAGGAATTGGCGGGGAGCACTACAACGGGTGGAGCCTGCGGTTTAATTGGATTCAACGCCGGGCATCTTACCA	<i>Methanococcus</i>
ACCGCCTGGGGAGTACGGCCGCAAGGTAAAACTCAAATGAATTGACGGGGGCCCGC•ACAAGCGGTGGAGCATGTGGTTTAATTCGATGCAACGCGAAGAACCTTACCT	<i>E. coli</i>
GTTCCGGGGGAGTATGGTTGCAAAGCTGAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human

Part of rRNA component of ribosome
(after 3.5 billion years of evolution)

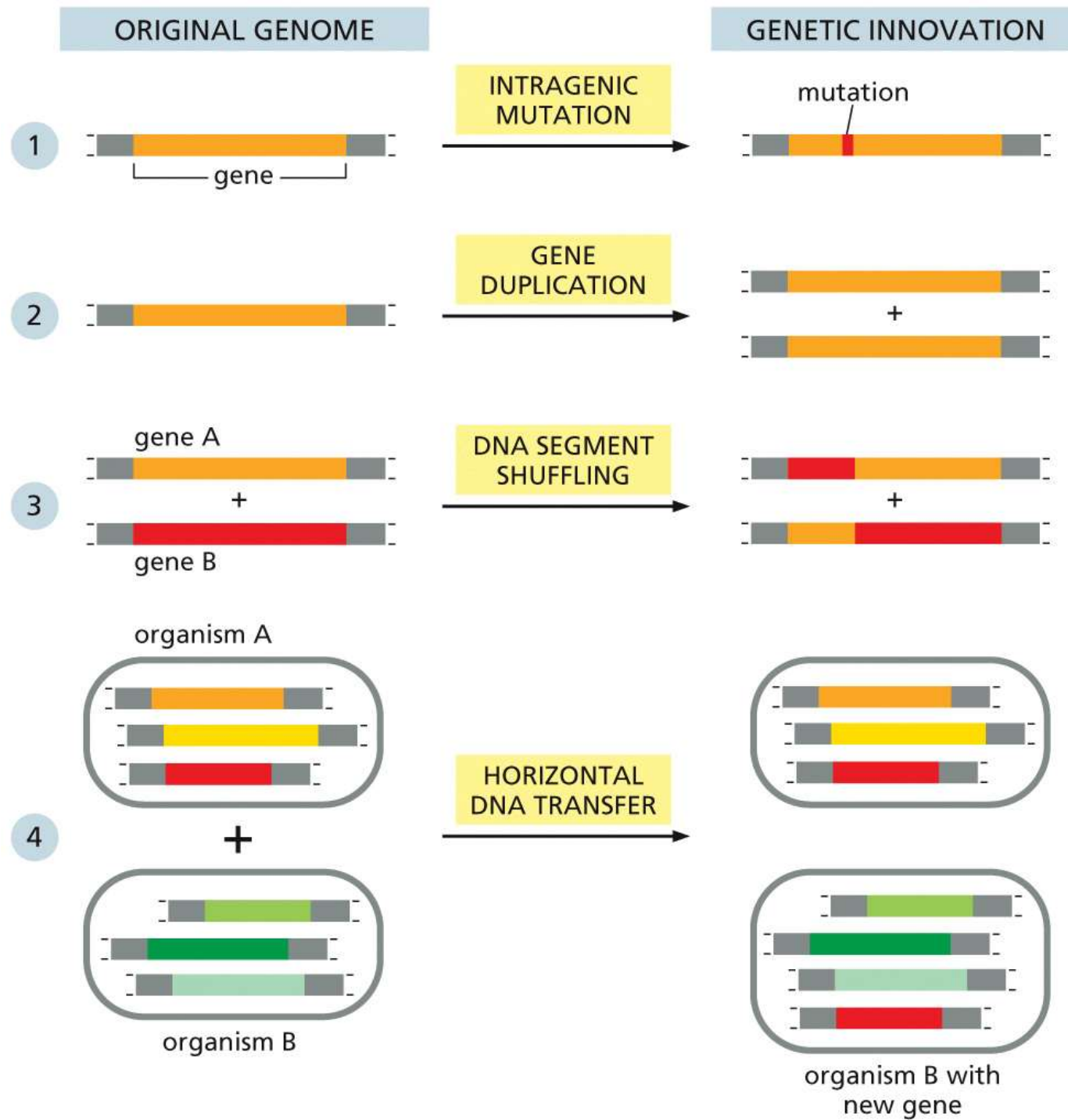
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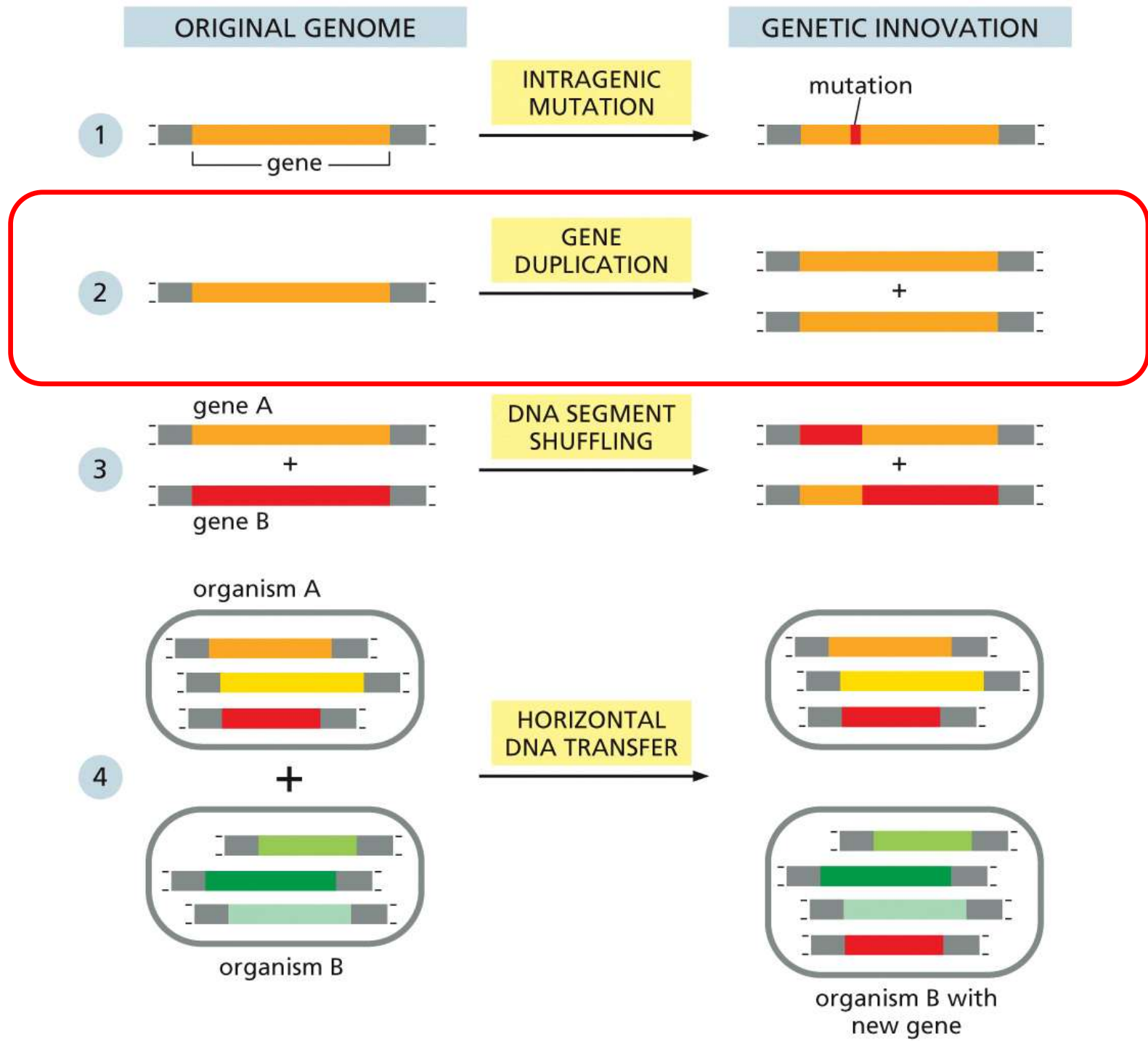
We?

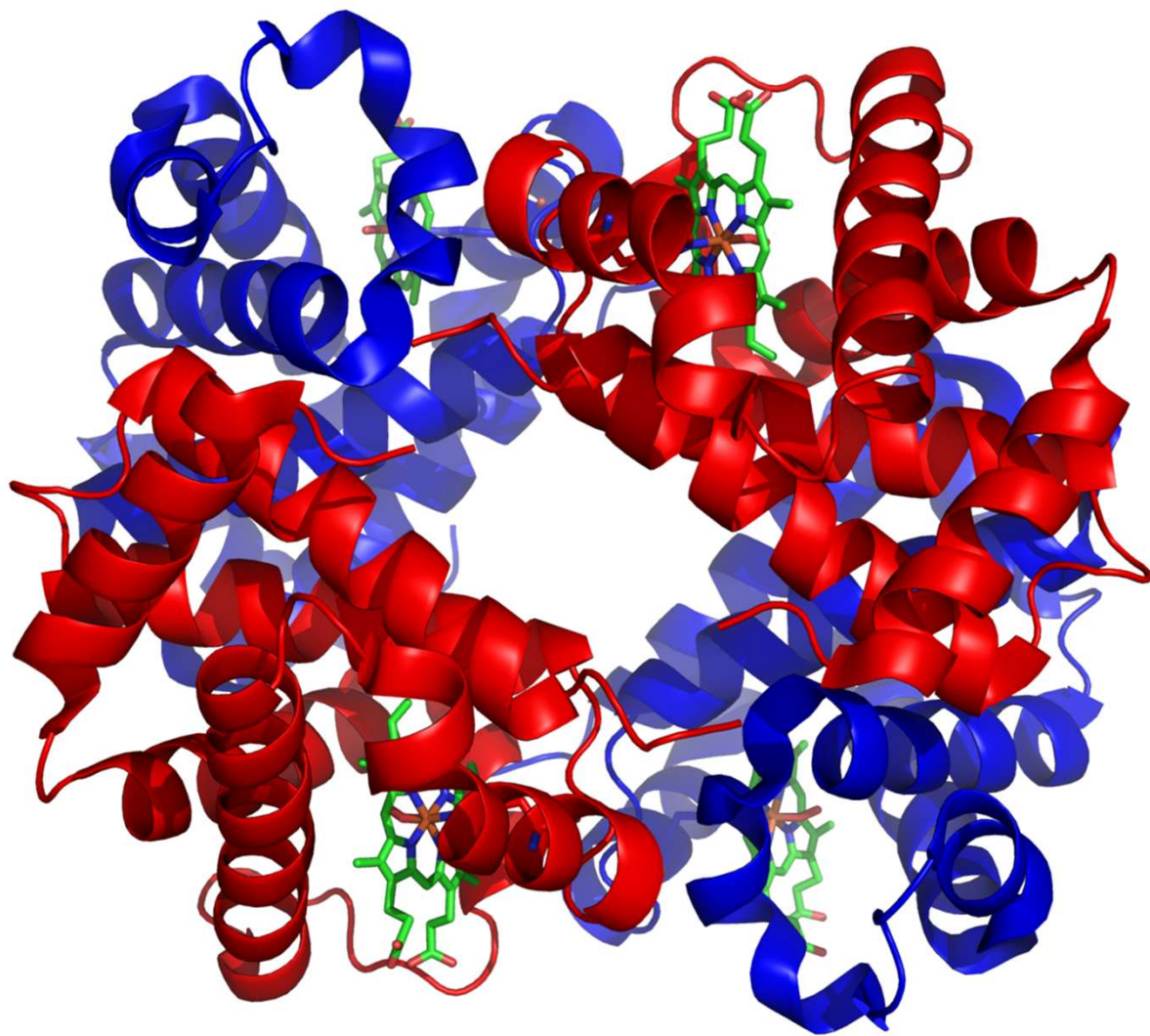
Where do 'new' genes come from?

Genome Diversification and the Tree of Life

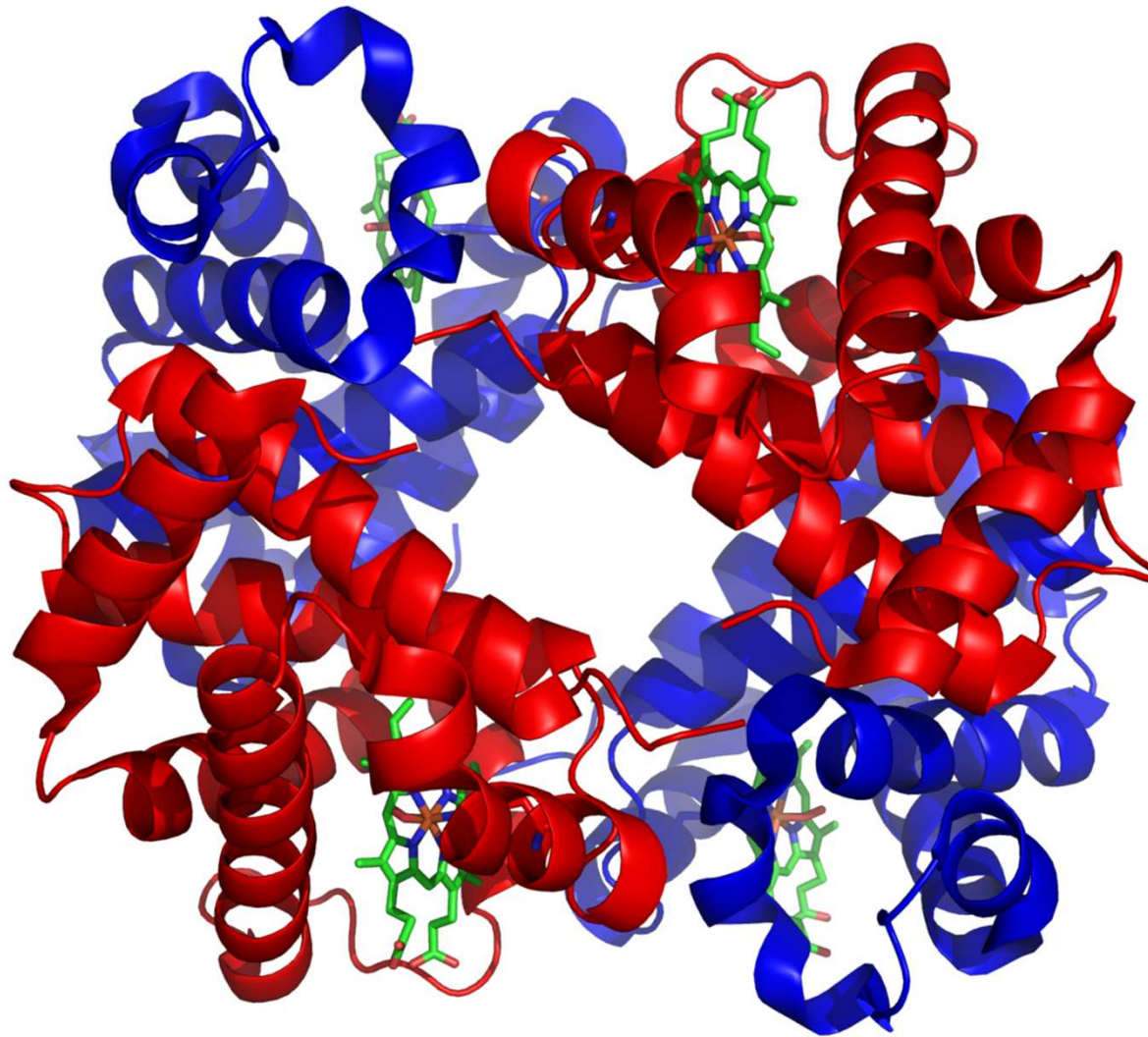
- New Genes Are Generated from Preexisting Genes



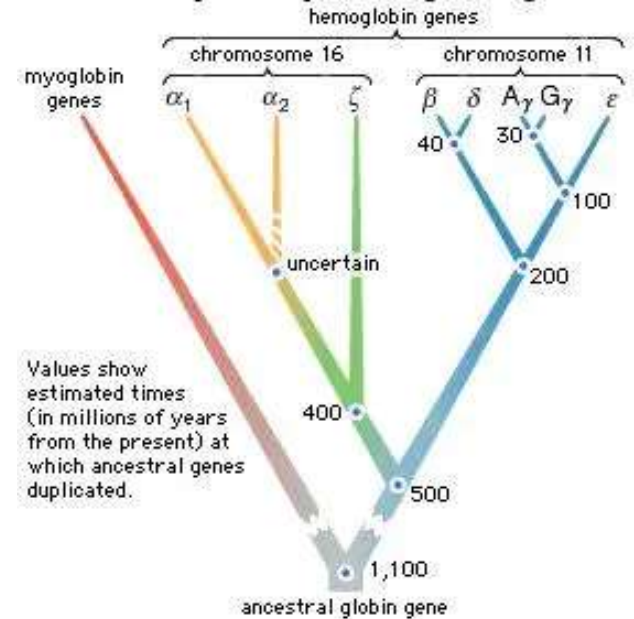




Example of gene duplication: the globin genes

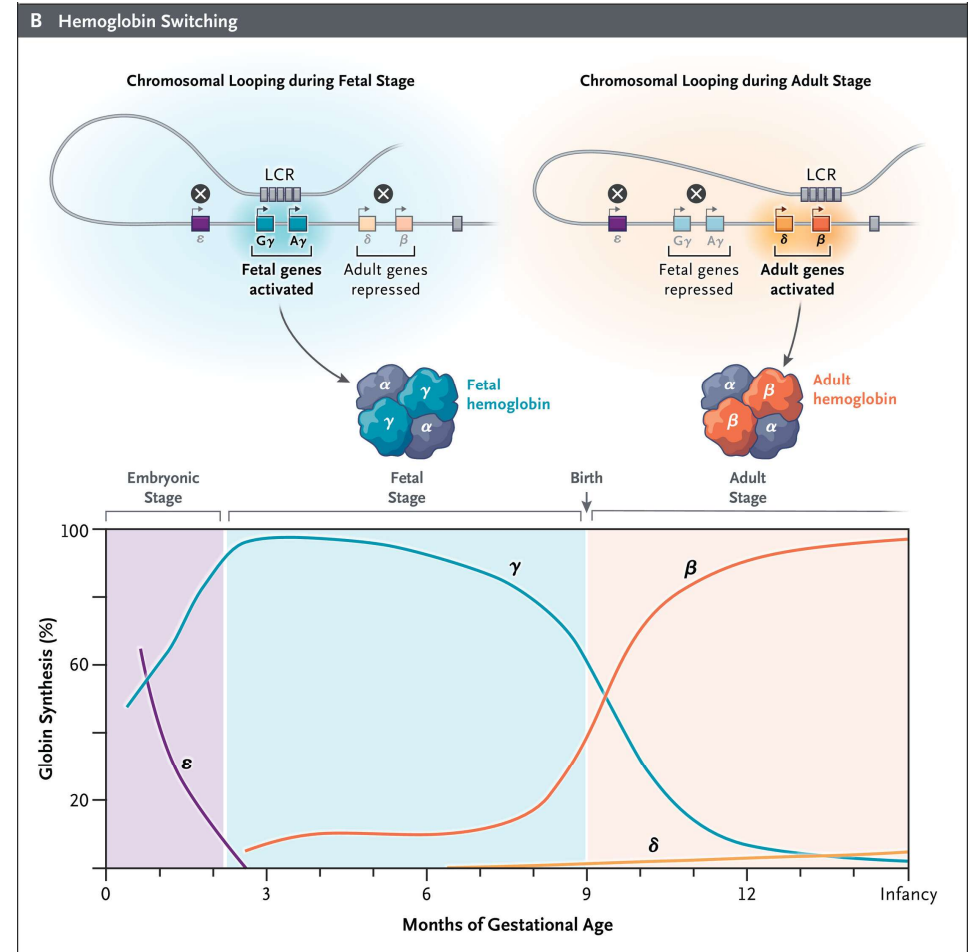
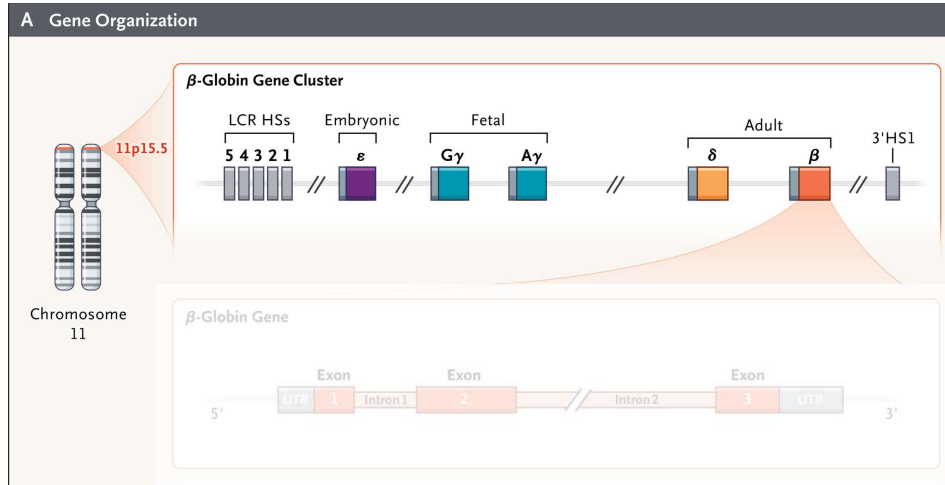


Evolutionary history of the globin genes



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Example of gene duplication: the globin genes



REVIEW ARTICLE | FRANKLIN H. EPSTEIN LECTURE

The Fetal-to-Adult Hemoglobin Switch — Mechanism and Therapy

This article has been corrected. [VIEW THE CORRECTION](#)

Author: Stuart H. Orkin, M.D. [Author Info & Affiliations](#)

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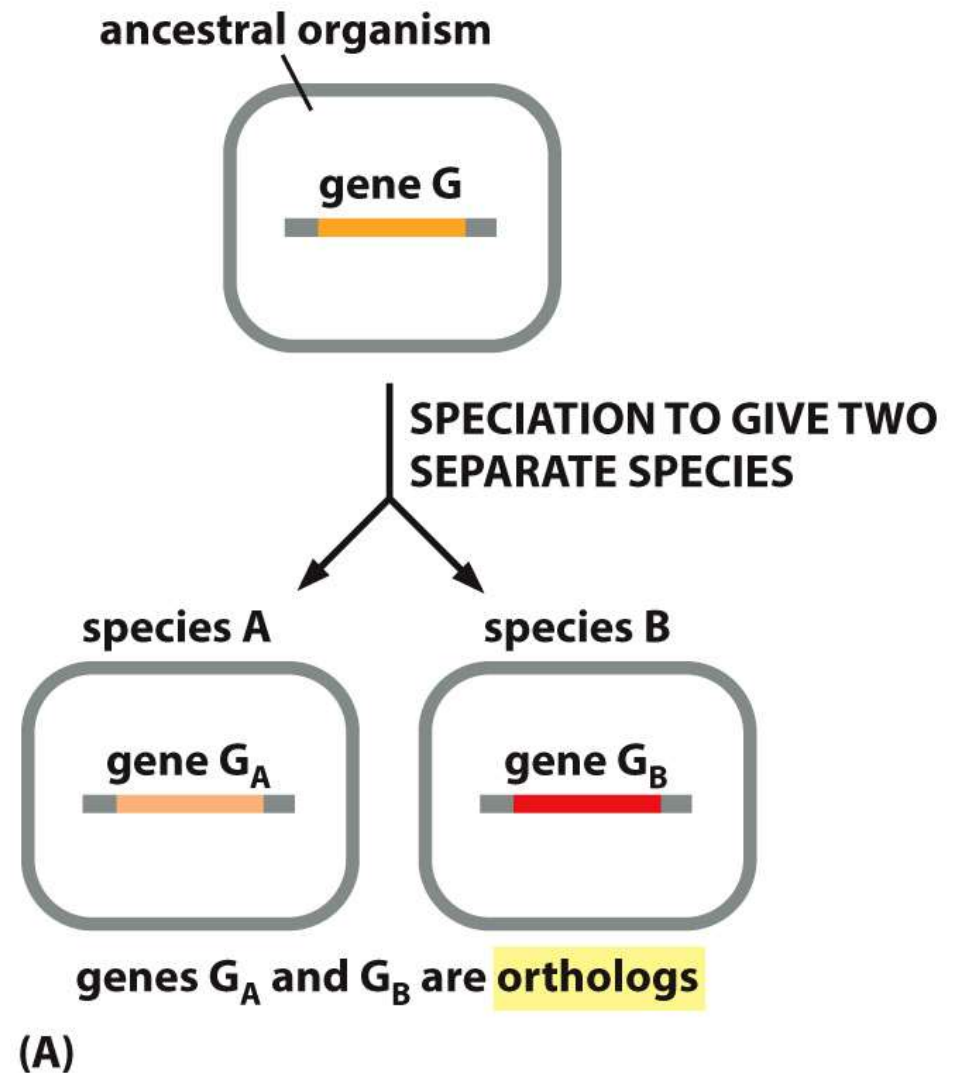
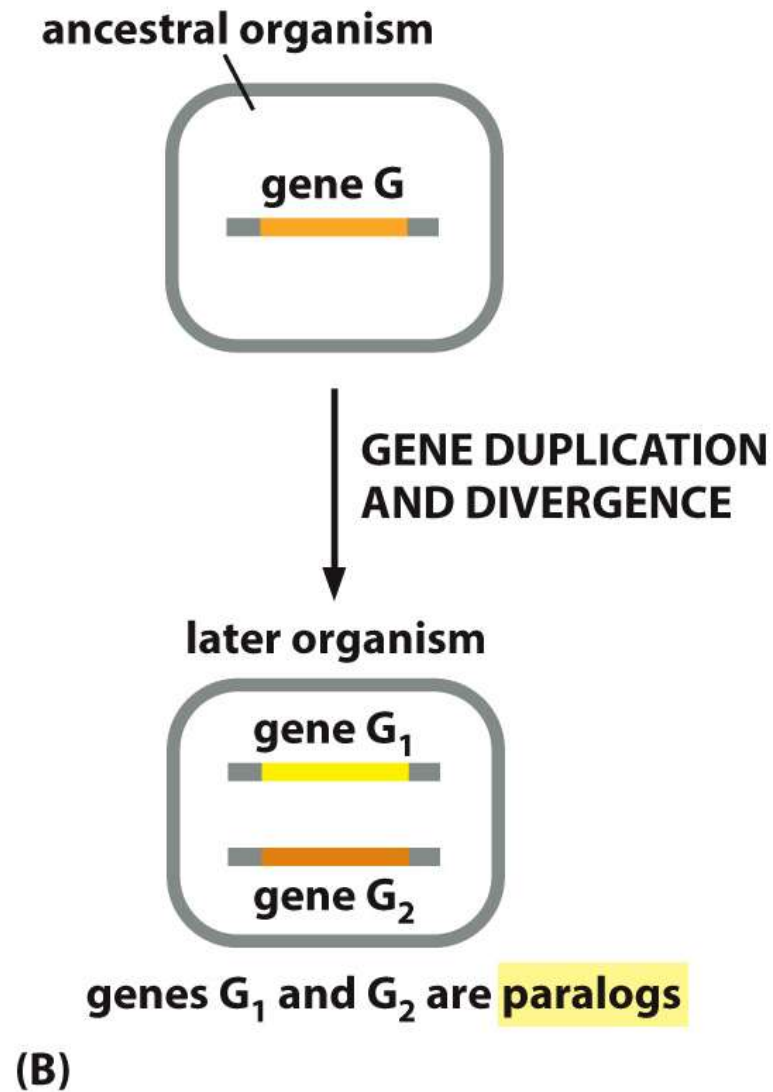


Figure 1-21 Molecular Biology of the Cell 6e (© Garland Science 2015)

Genome Diversification and the Tree of Life

- Gene Duplications Give Rise to Families of Related Genes Within a Single Genome

Genome Diversification and the Tree of Life

- Gene Duplications Give Rise to Families of Related Genes Within a Single Genome
- More Than 200 Gene Families Are Common to All Three Domains of Life

Family relationships

TABLE 1–1 The Number of Gene Families, Classified by Function, Common to All Three Domains of the Living World

Information processing		Metabolism	
Translation	63	Energy production and conversion	19
Transcription	7	Carbohydrate transport and metabolism	16
DNA replication, recombination, and repair	13	Amino acid transport and metabolism	43
Cellular processes and signaling		Nucleotide transport and metabolism	15
Cell-cycle control, mitosis, and meiosis	2	Coenzyme transport and metabolism	22
Defense mechanisms	3	Lipid transport and metabolism	9
Signal-transduction mechanisms	1	Inorganic ion transport and metabolism	8
Cell wall/membrane biogenesis	2	Secondary metabolite biosynthesis, transport, and catabolism	5
Intracellular trafficking and secretion	4	Poorly characterized	
Post-translational modification, protein turnover, chaperones	8	General biochemical function predicted; specific biological role unknown	24
For the purpose of this analysis, gene families are defined as “universal” if they are represented in the genomes of at least two diverse archaea (<i>Archaeoglobus fulgidus</i> and <i>Aeropyrum pernix</i>), two evolutionarily distant bacteria (<i>Escherichia coli</i> and <i>Bacillus subtilis</i>), and one eukaryote (yeast, <i>Saccharomyces cerevisiae</i>). (Data from R.L. Tatusov et al., <i>Science</i> 278:631–637, 1997; R.L. Tatusov et al., <i>BMC Bioinformatics</i> 4:41, 2003; and the COGs database at the US National Library of Medicine.)			

>200 gene families: deduce function of 'new' gene

Eukaryotic genomes are big (but large variations exist)

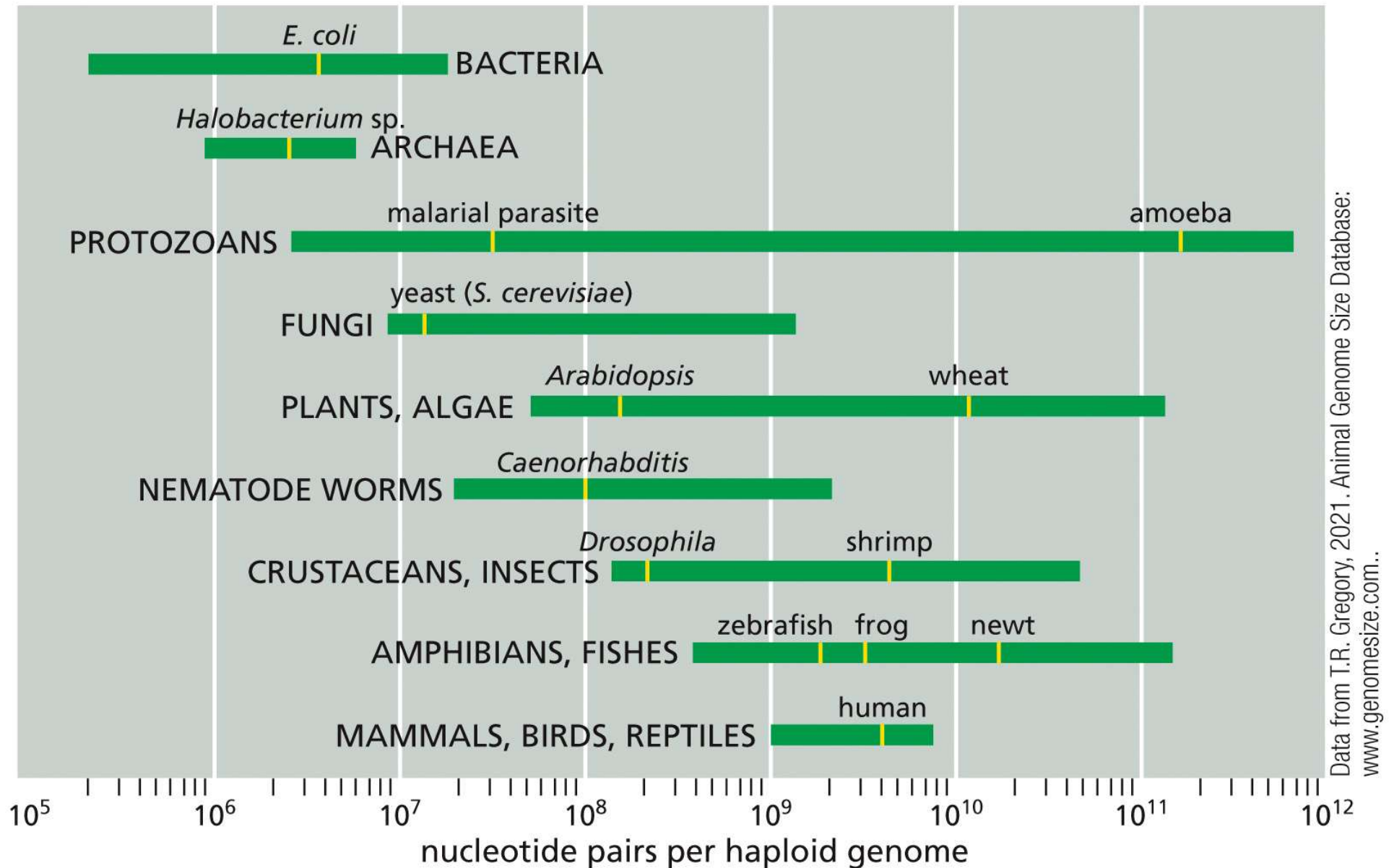


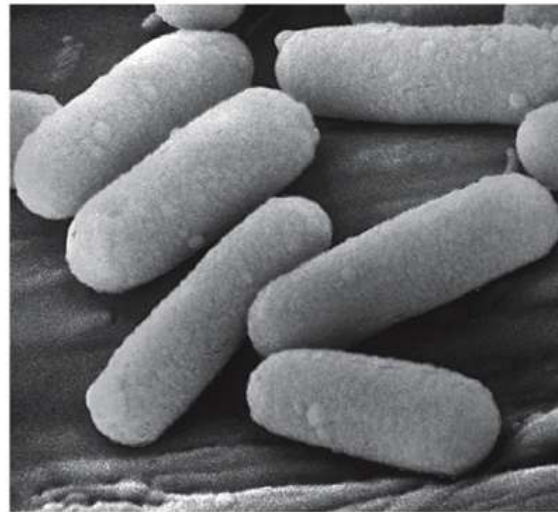
TABLE 1–2 Some Model Organisms and Their Genomes

Organism	Approximate genome size* (nucleotide pairs)	Approximate number of protein-coding genes**
<i>Escherichia coli</i> (bacterium)	4.6×10^6	4300
<i>Saccharomyces cerevisiae</i> (yeast)	12.5×10^6	6600
<i>Caenorhabditis elegans</i> (roundworm)	100×10^6	20,000
<i>Arabidopsis thaliana</i> (plant)	135×10^6	27,000
<i>Drosophila melanogaster</i> (fruit fly)	180×10^6	14,000
<i>Danio rerio</i> (zebrafish)	1400×10^6	26,000
<i>Mus musculus</i> (mouse)	2800×10^6	20,000
<i>Homo sapiens</i> (human)	3100×10^6	20,000

*Genome size includes an estimate for the amount of highly repeated, noncoding DNA sequence, which does not appear in genome databases.

**There are also genes that code for functional RNA molecules that do not code for proteins.

origin of replication



(A)

1 μm

Escherichia coli K-12
4,639,221 nucleotide pairs

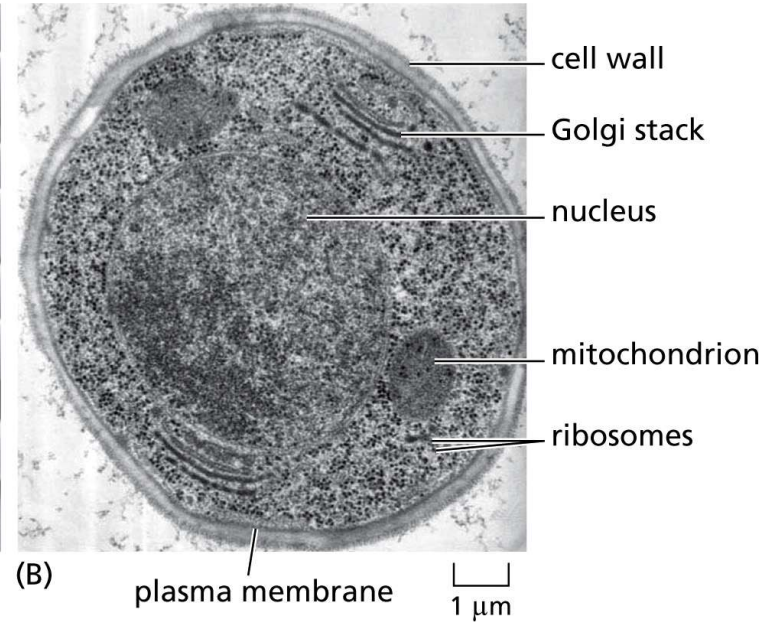
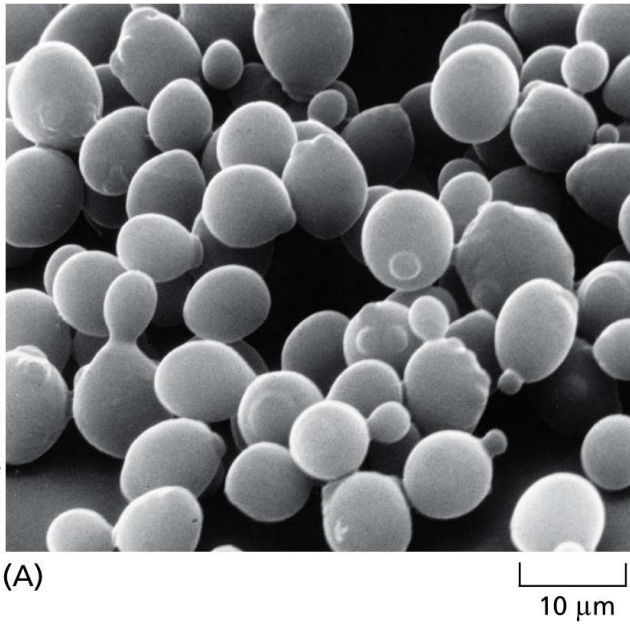
A, Dr. Tony Brain & David Parker/Science
Source; B, adapted from F.R. Blattner et al., *Science* 277:1453–1462, 1997.

terminus of replication



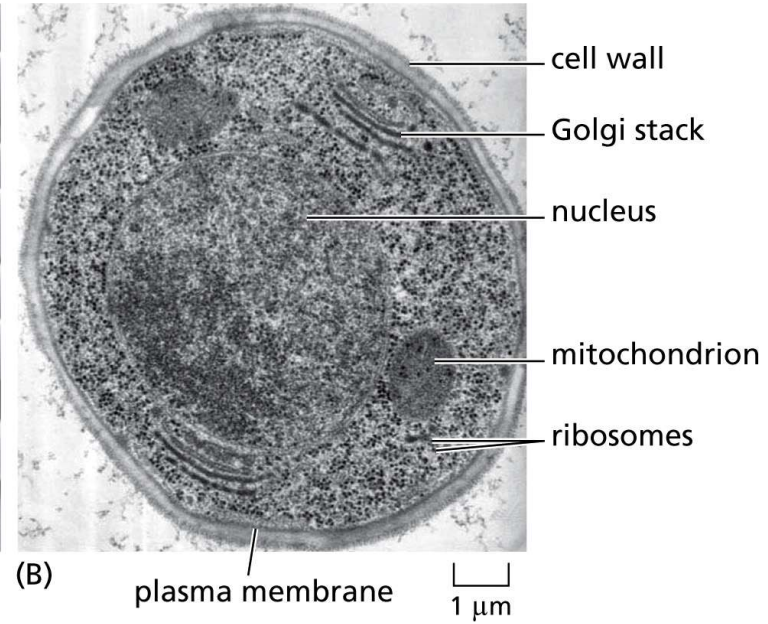
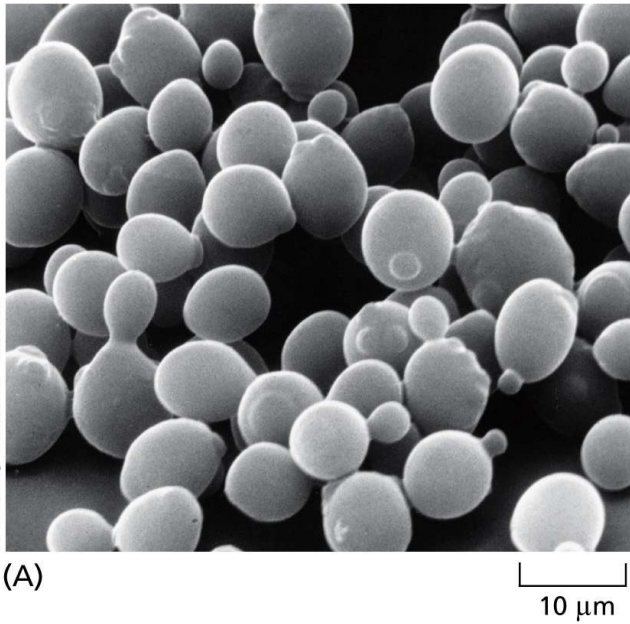
(B)

A, courtesy of Ira Herskowitz and Eric Schabach;
B, courtesy of Andrew Staehelin



Saccharomyces cerevisiae

A, courtesy of Ira Herskowitz and Eric Schabach;
B, courtesy of Andrew Staehelin



Saccharomyces cerevisiae



Courtesy of Maria Gallegos, University of Wisconsin, Madison

Caenorhabditis elegans

antennae



(A)

1 mm

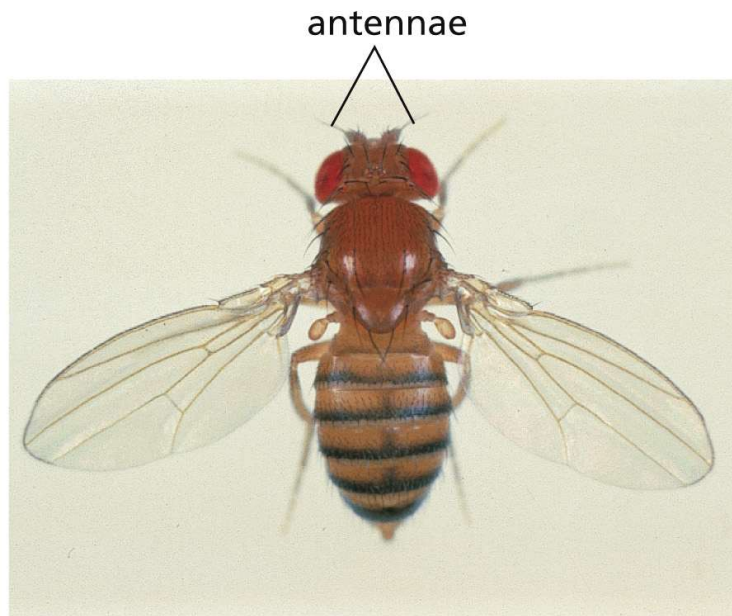
misplaced legs



(B)

A, Edward B. Lewis, Courtesy of the Archives, California
Institute of Technology; B, courtesy of Matthew Scott

Drosophila melanogaster



(A)

1 mm



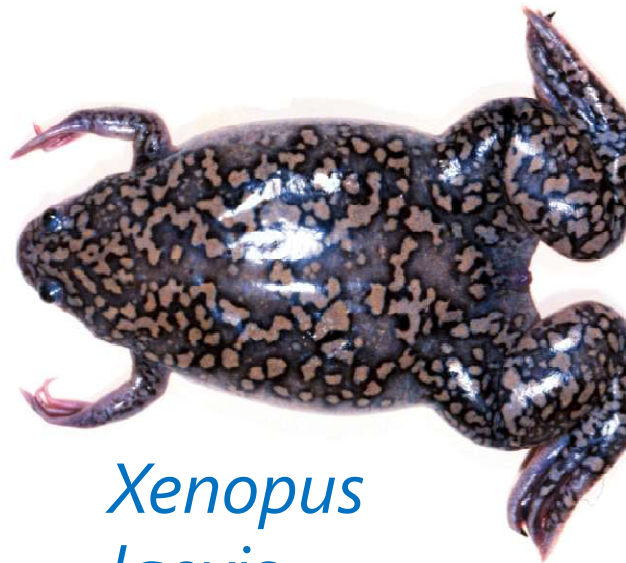
(B)

A, Edward B. Lewis, Courtesy of the Archives, California Institute of Technology; B, courtesy of Matthew Scott

Drosophila melanogaster



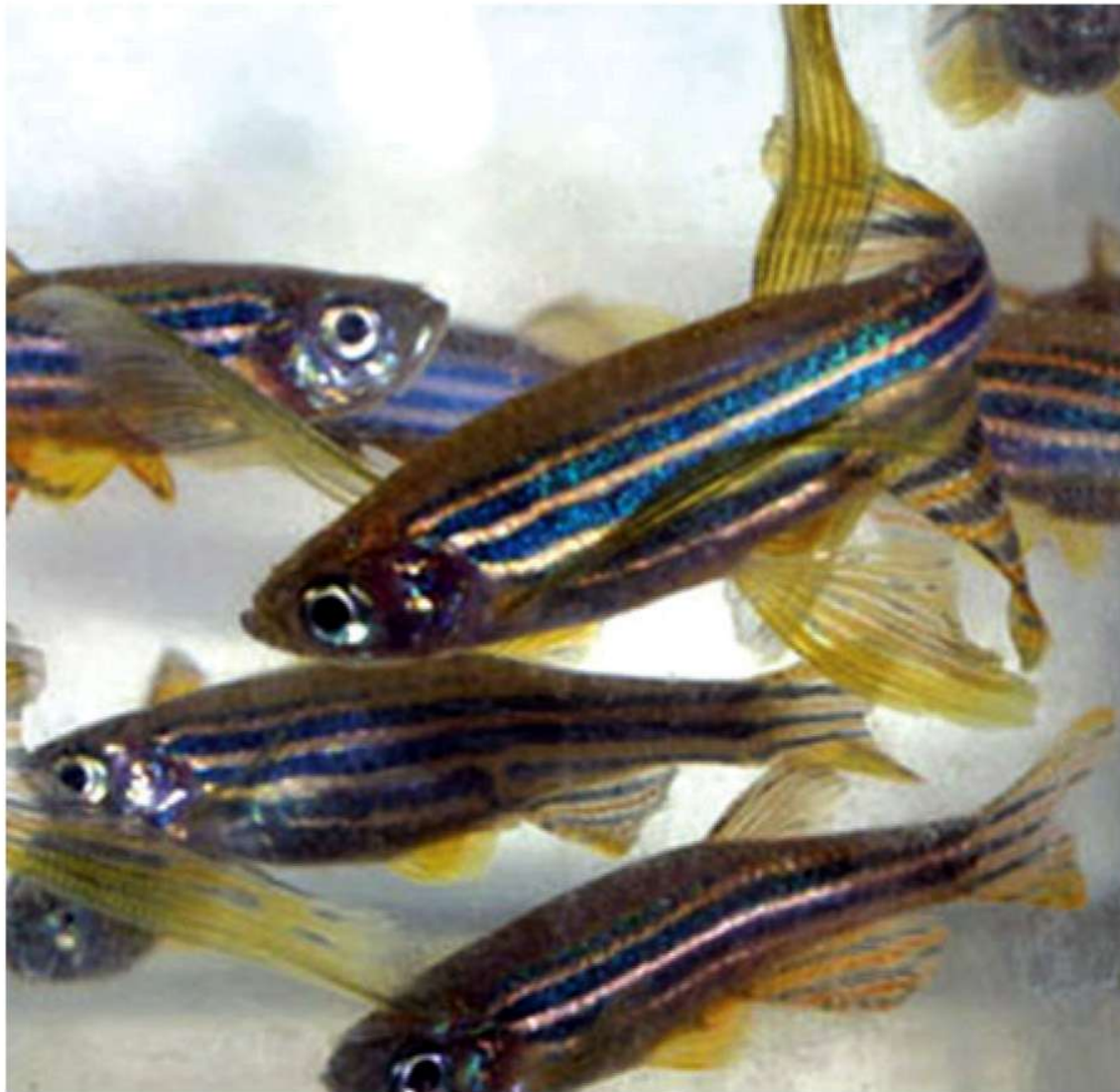
*Xenopus
tropicalis*



*Xenopus
laevis*

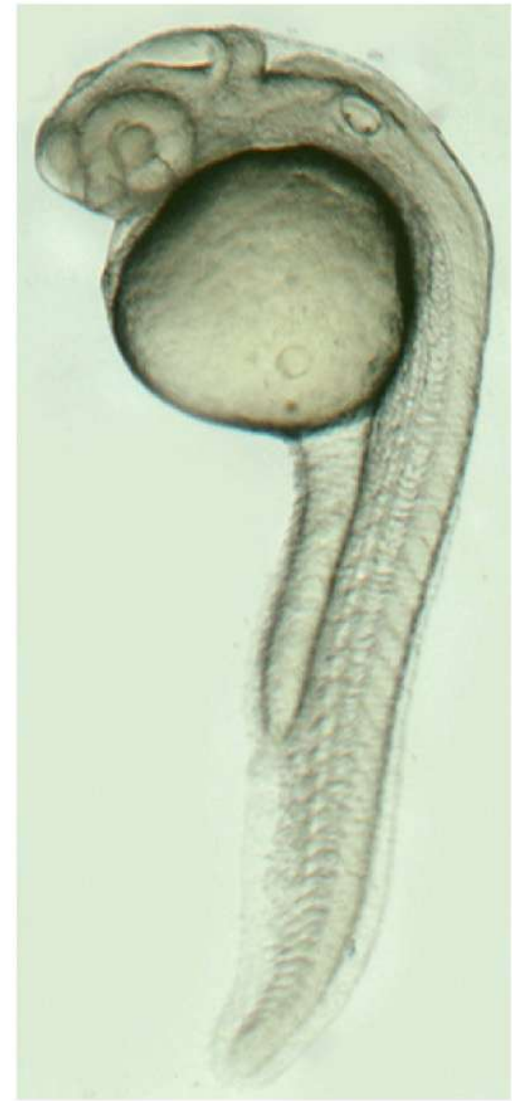
Figure 1-42. Molecular Biology of the Cell 6e (© Garland Science 2015)

Zebrafish



(A)

1 cm



(B)

150 μm